



MATHS in Extreme Sports

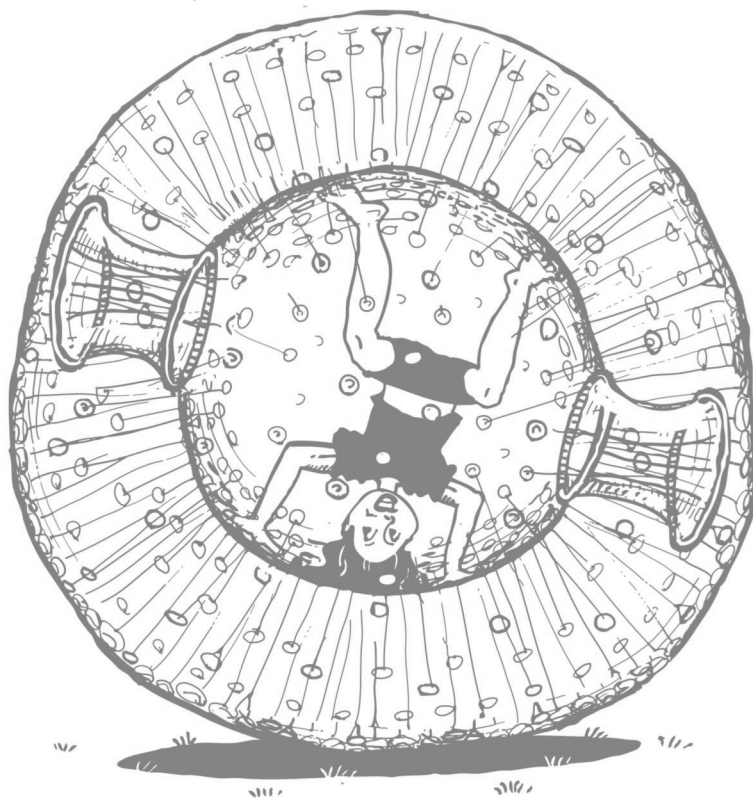


Simon Garner



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resources

MATHS in Extreme Sports



Simon Garner

This book was commissioned by Jennifer Gough, project management was by Piper Print Media, Gail McManus Graphics designed the cover and David Dickson supplied the illustrations.

Simon Garner is a secondary teacher with 15 years experience, a journalist and educational writer and producer. He is currently the Executive Producer of VEA, Australia's largest producer of educational DVDs for schools, technical and further education and the corporate sector.

Over the past decade Simon has specialised in educational writing, which has included scripting more than 200 nationally and internationally distributed educational video/DVD programs, a number of which he also produced; a range of extensive curriculum support materials for nationally accredited training programs; and a variety of consulting work for educational publishers. He is also the author of *Maths in Television* and the editor of the *Instant Lessons PowerPoint Geography* series.

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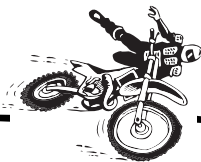
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Introduction



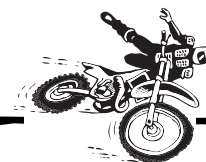
Maths in Extreme Sports contains photocopiable worksheets about some of the most challenging sports in which adventurous people around the world participate. Each topic has been written to interest and engage students in years 9–10, and contains a variety of content such as basic operations, measurement, graphs, estimation, conversions, percentages and analysis of data. The activities will also help students to see how the skills of maths can be used in everyday life.

Students may need to refer to an atlas for topic 3, Formula One racing. However, most students should need no more than their own general knowledge to complete the activities throughout this title. Some sports are featured over two topics, but do not have to be completed sequentially. Unless advised otherwise, students are asked to round their answers to two decimal places. For most topics, a calculator would be useful.

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Conversions - length, time



1

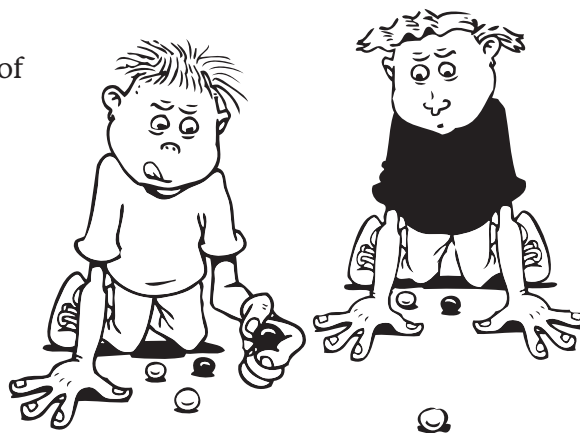
When you measure quantities such as distance, speed and mass, you need to use different units of measurement – depending on which quantity is being measured, and in which country the measurements are being made.

In Australia, metric units of measurement are used such as centimetres, metres, kilometres, kilometres per hour, grams and kilograms. In the USA, Imperial units of measurement such as feet, inches, miles, miles per hour, pounds and ounces are used. In Britain, metric units are often used for mass and some lengths, but road distance and road speeds are commonly measured in miles and miles per hour.

The size of the quantity you are measuring often determines the units to use. For example, the speed of a marble rolling along the floor would normally be measured in metres per second. Land speed records in a car would be measured in kilometres per hour.

So if the speed of a skater is measured in kilometres per hour in one place, and in miles per hour in another place, how do you compare them?

The secret is in converting units of measurement, so you can make meaningful comparisons.



Converting units of length or distance

The **metric system** of measurement makes it easy to convert units. It involves either **multiplying** or **dividing** numbers by tens, hundreds, thousands and so on. For quantities such as length and mass, conversions can be made by simply moving the position of the decimal point in numbers an appropriate number of places in the correct direction.

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

1 Sprint your way through the following metric conversions.

Remember:

- when you are converting from a **smaller unit to a bigger unit**, the appropriate number or factor should be **divided**
- when you are converting from a **bigger unit to a smaller unit**, the appropriate number or factor should be **multiplied**.

a 1.6 kilometres (km) = _____ metres (m)

b 256 millimetres (mm) = _____ centimetres (cm) [to 1 decimal place]

c 14 m = _____ cm

d 765 m = _____ km (to 3 decimal places)

Imperial units of length include inches, feet, yards and miles. Conversion between units does not involve multiplying or dividing numbers by tens, hundreds or thousands. Some examples are shown in the box at the right.

When converting from Imperial units of measurement to metric units and vice versa, it is helpful to know the conversion factors – that is, the numbers by which you multiply or divide to correctly convert the units. Some examples are shown in the box at the right.

Imperial units of length

12 inches (in) = 1 foot (ft)

3 feet (ft) = 1 yard (yd)

5280 ft = 1 mile (mi)

1760 yd = 1 mi

1 in = 2.54 cm

1 ft = 30.48 cm

1 ft = 0.31 m

1 mi = 1609.34 m

1 mi = 1.61 km

1 yd = 91.44 cm

Conversions - length, time (cont'd)



1

Now bolt your way through the following conversions.

Remember:

- when you are converting from **a smaller unit to a bigger unit**, the appropriate number or factor should be **divided**
- when you are converting from **a bigger unit to a smaller unit**, the appropriate number or factor should be **multiplied**.

2 Convert these Imperial measurements to metric measurements.

- a** 320 ft = _____ m **b** 67.8 mi = _____ km **c** 45 yd = _____ cm
d 7.38 mi = _____ m **e** 8.5 in = _____ cm **f** 0.75 yd = _____ cm

3 Convert these metric measurements to Imperial measurements.

- a** 91.69 cm = _____ ft **b** 89 km = _____ mi **c** 24.91 cm = _____ yd
d 19 538 m = _____ mi **e** 17.95 cm = _____ in **f** 4.62 km = _____ mi

Converting units of time

Time is measured by the same units everywhere, and these units include seconds, minutes and hours. When comparing or converting time units from one to another, it is helpful to know the conversion factors which are shown at the right.

Time conversions

1 minute (min) = 60 seconds (s)
 1 hour (h) = 60 minutes (min)
 1 hour (h) = 3 600 seconds (s)

4 Aim for a personal best, as you work through these time conversions:

- a** 27 min = _____ h **b** 8.45 h = _____ min
c 46 791 s = _____ min = _____ h **d** 7844 s = _____ h
e 2.45 h = _____ min = _____ s **f** 0.78 h = _____ s
g 848 min = _____ s **h** 72 000 s = _____ h

When converting time, it is important to understand when and how to convert decimal components of the units when you are using a combination of hours, minutes and seconds in your answer.

For example:

What is 3.89 hours in hours and minutes?

First convert 0.89 hours to minutes:

$$0.89 \times 60 = 53.4 \text{ min}$$

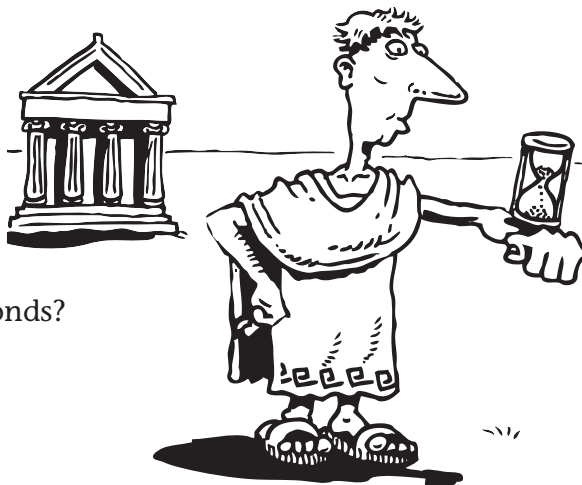
So 3.89 h = 3 h 53.4 min

What is this answer in hours, minutes and seconds?

Convert 0.4 min to seconds:

$$0.4 \times 60 = 24 \text{ s}$$

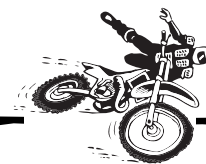
So 3.89 h = 3 h 53 min 24 sec



5 Take on the challenge and complete these time conversions.

- a** 17.11 h = _____ min **b** 17.11 h = _____ h _____ min
c 17.11 h = _____ h _____ min _____ s **d** 46.34 min = _____ s
e 46.24 min = _____ min _____ s **f** 0.58h = _____ min _____ s

Conversions - speed, mass, volume



2

When you measure quantities such as distance, speed and mass, you need to use different units of measurement – depending on which quantity is being measured, and in which country the measurements are being made.

In Australia, metric units of measurement are used such as centimetres, metres, kilometres, kilometres per hour, grams and kilograms. In the USA, Imperial units of measurement such as feet, inches, miles, miles per hour, pounds and ounces are used. In Britain, metric units are often used for mass and some lengths, but road distance and road speeds are commonly measured in miles and miles per hour.

Converting units of speed

A unit of speed is always shown as a unit of length or distance divided by (or 'per') a unit of time. Metric units of length or distance include centimetres, metres and kilometres; and Imperial units include feet, yards and miles.

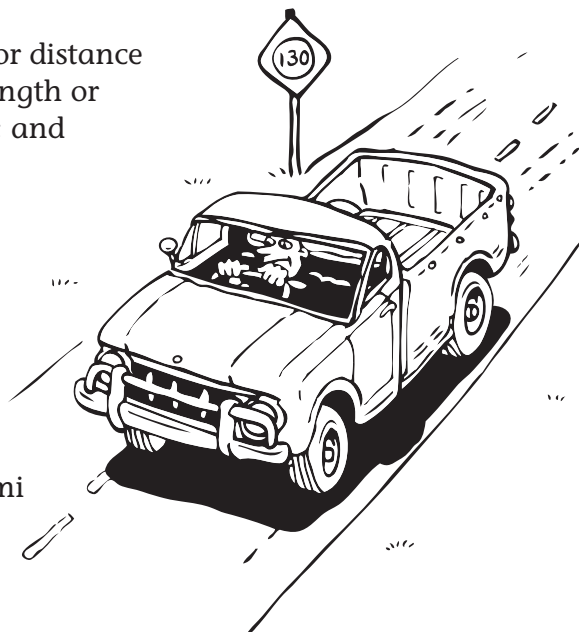
It's time to look at converting units of speed. Let's start with converting kilometres per hour (km/h) to miles per hour (mph or mi/h).

For example, the maximum speed limit in Australia is 130 km/h on the open highway in the Northern Territory. What is this speed in mph?

Converting km to mi: $130 \text{ km} = 130 \div 1.61 = 80.75 \text{ mi}$

In both units of speed, the time unit is the same:
h (hours).

So $130 \text{ km/h} = 80.75 \text{ mph}$



1 Complete the sentences:

- a To convert km/h to mph, we _____ the km/h figure by 1.61.
- b To convert mph to km/h, we _____ the mph figure by 1.61.

- 2
- a $150 \text{ km/h} = \text{_____ mph}$
 - b $32 \text{ mph} = \text{_____ km/h}$

What happens when both the units of distance and the units of time need to be converted? Two examples are shown below.

How do we convert 18.6 km/h to m/s?

First convert the unit of length or distance: $18.6 \text{ km} = 18\,600 \text{ m}$
Then convert the unit of time: $1 \text{ h} = 3\,600 \text{ s}$
We now have an equation to solve: $18\,600 \text{ m} \div 3\,600 \text{ s} = 5.17 \text{ m/s}$
(Rounded to 2 decimal places)

How do we convert 50 mph to m/s?

First convert the unit of length or distance: $50 \text{ mi} \times 1\,609.34 = 80\,647$
Then convert the unit of time: $1 \text{ h} = 3\,600 \text{ s}$
The equation: $80\,647 \div 3\,600 = 22.40 \text{ m/s}$



3 Try these speed conversions!

a Convert 8.5 mph to m/s.

b Convert 1.53 m/s to km/h.

c Convert 419.23 m/s to mph.

d Convert 390 km/h to m/s.

Converting units of mass

4 Answer these questions about metric units of mass.

a How many grams (g) are there in 1 kilogram (kg)?

b How many kilograms (kg) are there in 1 tonne or metric ton (t)?

c How many grams (g) are there in 1 tonne (t)?

Two Imperial units of mass are **ounces (oz)** and **pounds (lb)**:

1 lb = 16 oz	1 lb = 453.59 g
1 oz = 28.35 g	1 lb = 0.45 kg

5 Convert the following Imperial measurements to metric measurements.

a 18.76 oz = _____ g

b 0.92 lb = _____ g

c 388 lb = _____ kg

d 14 lb 6 oz = _____ kg

6 Convert the following metric measurements to Imperial measurements.

a 199.9 g = _____ oz

b 8 459 g = _____ lb

c 68.3 kg = _____ lb

d 19 kg = _____ lb _____ oz (Round your answer to the nearest full ounce.)

Did you know?

People often confuse **mass** and **weight**, thinking they are the same thing. In fact, they are different.

Mass refers to how much matter is contained in something. It is measured in metric units such as grams, kilograms and tons, and Imperial units such as ounces and pounds.

Weight refers to the force with which gravity pulls downwards on an object or living thing. It is measured in newtons (N), which is the unit of measurement for force.

Converting units of liquid volume

Units of measurement of liquid volume also vary between metric and Imperial systems. Here we will look at just two units – metric litres (L) and Imperial gallons (gal).

$$3.79 \text{ L} = 1 \text{ gal}$$

7 If a fuel tank holds 90 L, how many gallons is this?

8 It takes 659 630.6 gal of water to fill an Olympic-sized swimming pool. How many litres is this? (Round your answer to the nearest litre.)

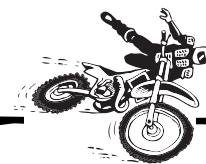
9 A container with a volume of 1 000 cm³ will hold exactly 1 L of water. If a cube has equal sides of 0.46 m, how many litres of water would it hold?

10 A fire-fighting helicopter can drop 4 000 L of water in one dump.

a What is the volume of the water reservoir on the helicopter, in cm³?

b How many gallons of water does it hold?



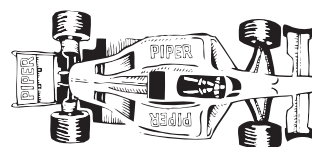


Formula One car racing ranks as one of the world's fastest, potentially most dangerous, and most popular sports. Hundreds of thousands of fans flock to circuits around the globe to watch the events, with hundreds of millions more watching on television. Little wonder – these vehicles travel at wild speeds. Formula One combines cutting edge racing engineering with the steely skills of the world's top drivers as they push their amazingly quick machines to speeds upwards of 300 km/h.

Tip: An atlas will be useful for completing this unit!

In 2008, the following circuits around the world hosted Formula One car races:

-  **Albert Park Melbourne, Australia** – street circuit
-  **Bahrain International Circuit, Bahrain** – race circuit
-  **Catalunya, Barcelona** – race circuit
-  **Fuji Speedway, Japan** – race circuit
-  **Villeneuve Circuit, Montreal, Canada** – road/race circuit
-  **Hockenheim, Germany** – road/race circuit
-  **Hungaroring, Budapest** – race circuit
-  **Istanbul Park, Istanbul** – race circuit
-  **Autodromo Jose Carlos Pace, Sao Paulo, Brazil** – race circuit
-  **Magny-Cours Circuit, France** – race circuit
-  **Monte Carlo Circuit, Monaco** – street circuit
-  **Autodromo Nazionale, Monza, Italy** – race circuit
-  **Sepang International Circuit, Kuala Lumpur, Malaysia** – race circuit
-  **Shanghai International Circuit, Shanghai, China** – race circuit
-  **Silverstone Circuit, Britain** – race circuit
-  **Singapore Circuit** – street circuit
-  **Spa Francorchamps, Belgium** – street circuit
-  **Valencia Circuit, Spain** – street circuit



Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

- 1 What do you think is the difference between a **street circuit** and a **race circuit**?
- 2 What **percentage** of the 18 circuits worldwide are street circuits?
- 3 What **proportion** of the 18 events were held in the northern hemisphere?
- 4 What is the simplest ratio of Formula One Grand Prix races held in the northern hemisphere to those held in the southern hemisphere?

Of the circuits listed, the Spa Francorchamps circuit in Belgium is the longest at 7.004 km. The shortest is Monte Carlo, at just 3.34 km long.

- 5 To 3 decimal places, how much longer is the Belgian circuit than the Monte Carlo circuit?
- 6 Answer the following to 3 decimal places. If the average length of the 18 circuits is 5.05 km:
 - a how much shorter than the average circuit length is the Monte Carlo circuit?
 - b how much longer than the average circuit length is the Belgian circuit?
 - c how much shorter than the average circuit length is the Brazilian circuit, which is 4.309 km?

- 7
- The lap record on the Brazilian circuit was set in 2004 by Rubens Barichello in a Ferrari.
- a

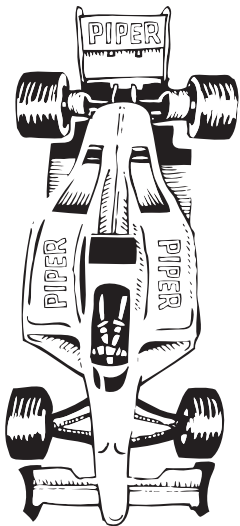
Barichello completed a single lap in 1.10.229. The 1 refers to minutes and the 10 to seconds. What do the remaining numbers refer to?
- b

The next fastest lap time was 1.11.114. By how much did this driver's time exceed Barichello's?
- c

What was Barichello's average speed in km/h on his fastest lap?
Hint: When converting minutes to hours, use the answer to 5 decimal places. Give the average speed to 2 decimal places.
- d

What was the second fastest driver's average speed in km/h on his fastest lap? *Hint:* When converting minutes to hours, use the answer to 4 decimal places. Give the average speed to 2 decimal places.
- e

How much quicker was Barichello's average speed on his fastest lap than the second fastest driver's average speed?



- 8
- In the 'Formula One Point Five' car racing circuit, drivers from different teams compete for points that tally up over the course of a year to choose a champion driver. The table below sums up last year's competition results. The teams refer to the manufacturers of the vehicles, the points refer to the collective tally gained over the racing season and the circuit refers to the race track at which each driver drove his fastest lap.
- In random order:
- the drivers were Lewis Webber, Felipe Rosberg, Rubens Hamilton, Mark Massa and Nico Barichello

•

the points scored by the drivers were: 3 814, 3 389, 3 406, 4 121 and 3 752

•

the circuits where the fastest laps were recorded were: Japan, San Marino, China, Melbourne and Brazil.

Complete the table using the following information.

Team	Driver	Points	Circuit
Wilkins			
Hando			
Fer-rar-ree			
Blue Cow			
McKenzie			

- Nico Barichello scored more points than any other driver, except the Hando team driver.

•

Mark Massa's fastest lap time was not recorded in Asia, nor was that of the Wilkins' team driver.

•

Lewis Webber, Rubens Hamilton and Felipe Rosberg all started their careers with the McKenzie team, but have now moved to other teams.
- The driver with the most points scored his fastest lap time in Melbourne.

•

Blue Cow team has always dominated the Brazil circuit, and this year was no exception.

•

Rubens Hamilton's score of 3 406 was his best tally in five years, although he did lose points in one race for running the Blue Cow driver off the track.
- Wilkins' driver scored fewer points than the Blue Cow team driver, but more points than Lewis Webber.

•

This year was the first time Mark Massa has topped the points table, and the first time that the Blue Cow driver finished third.

•

This year the Fer-rar-ree driver set a personal best lap time, which was also a new record for the Japanese circuit.



Game fishing is fishing at the serious end! This sport involves fishing for large ocean fish with a mass of 400 kg or more. It is popular in many parts of Australia, including along the north coast of Queensland. Species of game fish found in that region include Black marlin, Sailfish, Yellow fin tuna, Mackerel and Wahoo. Many companies offer fishing charters where groups of people can hire out a boat and crew and try their hand at this extreme sport.

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

Two couples – Anastasia and Brad, and Omar and Sally – go on a game fishing charter trip for a day aboard the vessel *The Monster*. It leaves Port Douglas in North Queensland at 9 am on a Saturday morning.

The Monster is powered by two 440 horsepower (hp) diesel engines. Horsepower is a unit of measurement of power used in relation to engines. Its use is not as common now as it once was. Today, the most common unit of measurement of power is Watts (W). When measuring engine power, 1 hp is approximately equal to 746W.

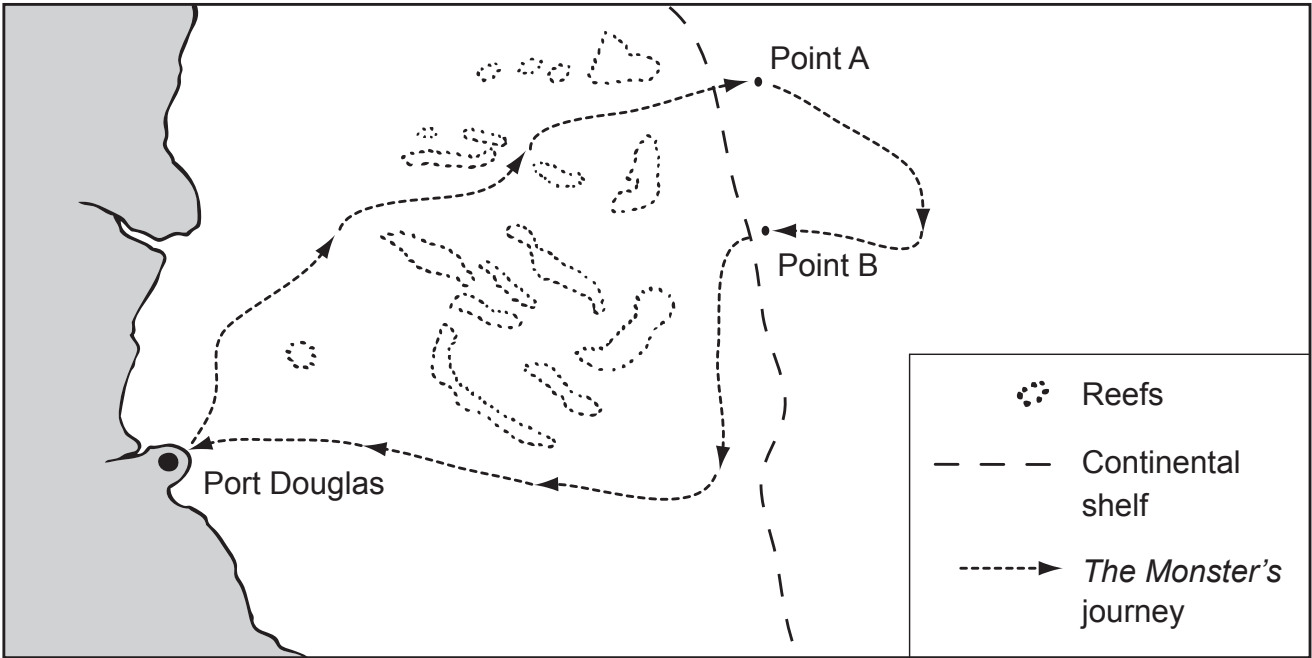
1 What is the maximum power output of *The Monster's* two engines in watts?

At sea, distances can be measured in nautical miles (n miles). This unit is different from miles, the Imperial unit of measurement for road distances. The speed at which a boat travels across the surface of the sea is often measured in knots (kn).

- 1 kn is equal to 1 n mile per hour or 1 n mile/h
- Speed can also be measured in the metric units of kilometres per hour (km/h)
- 1 kn is equivalent to 1.88 km/h

2 a If a boat is travelling at 19 kn, what is this speed in km/h?
b If a boat is travelling at 64.5 km/h, what is this speed in kn?

Back on our charter, plenty of fishing action went on while the boat was in motion, but *The Monster* also stopped at two different spots during the day where the engines were turned off. These spots were both located beyond a number of reefs – just past the Continental Shelf, which lies about 55 km off the coast of Port Douglas.

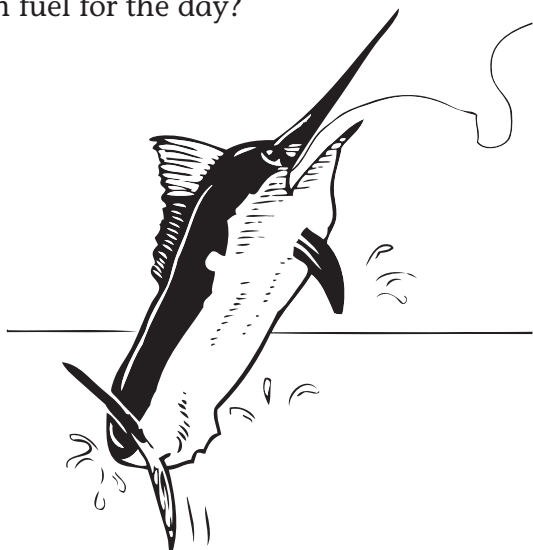
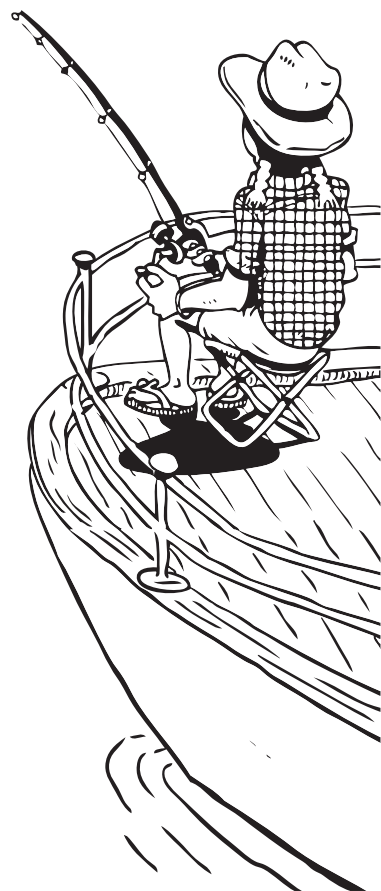


Game fishing - ocean travel (cont'd)



4

- 3 The distance between Port Douglas and Point A is 61.9 km.
 - a If *The Monster* travelled this distance in 1.4 hours, what was its average speed in km/h?
 - b What is this average speed in kn?
- 4 *The Monster* cruised its way around and got from Point A to Point B in a total of 90 minutes, travelling at an average speed of 39 km/h. What distance in kilometres did *The Monster* travel between Point A and Point B?
- 5 During the journey home from Point B to Port Douglas, *The Monster* cruised at an average speed of 27 kn.
 - a What is this speed in km/h?
 - b If the total distance travelled between Point B and Port Douglas was 94.8 km, how long in hours did the journey take?
 - c What is this time in hours and minutes? (Answer to the nearest minute.)
- 6 The boat stopped at Point A for 55 minutes and Point B for 46 minutes.
 - a Calculate how long in hours the group was on the water during the charter.
 - b What is this time in hours and minutes? (Answer to the nearest minute.)
- 7 *The Monster* left Port Douglas with two full fuel tanks, each holding 60 litres of diesel fuel.
 - a During the charter, it used up 58% of its fuel. How many litres of fuel was that?
 - b If the owners of *The Monster* paid \$1.91 per litre of diesel fuel, what was the cost of the fuel used on the charter?
- 8 Each person on the charter paid \$220 for the day.
 - a What were the total takings for the day?
 - b What proportion of the total takings was spent on fuel for the day?
- 9 The total cost of the other items needed to take the two couples out for the day was \$228.54. Each of *The Monster's* two crew members was paid \$200 for the day. How much profit did the owners of *The Monster* make on that particular day?
- 10 If *The Monster* takes a total of 32 people out in a week, each paying \$220, and total expenses are \$5 167.80, what is the profit made for that week?

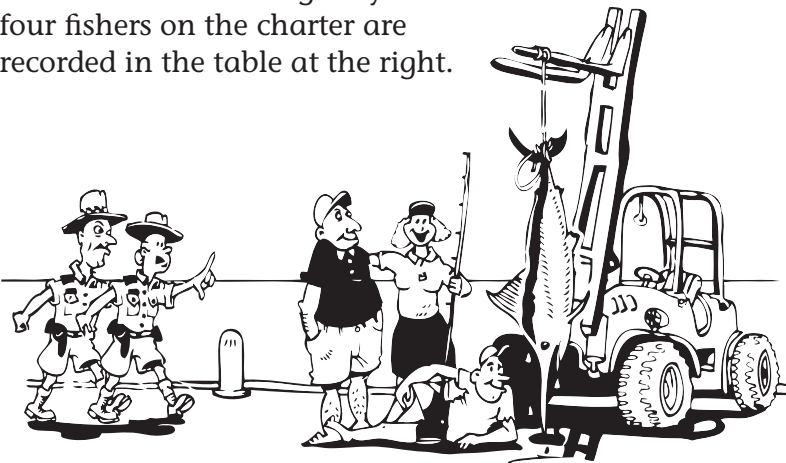




Anastasia and Brad and Omar and Sally are out on a game fishing charter trip aboard the vessel 'The Monster'. They head out from Port Douglas in North Queensland to try their hand at this extreme sport as they search for Black marlin, Sailfish, Yellow fin tuna, Mackerel and Wahoo - all tag and release, of course!

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

Details of the fish caught by the four fishers on the charter are recorded in the table at the right.



- 1 Rank the fishers according to the number of fish each one caught, from highest to lowest.
- 2 Rank the fishers according to the combined mass of fish each person caught, from highest to lowest.
- 3 Graph the information for questions 1 and 2, using the column graph format.
 - The first graph will have each person along the x-axis (horizontal axis), and the number of fish caught along the y-axis (vertical axis).
 - The second graph will have each person along the x-axis, and the combined mass of the fish they caught along the vertical y-axis.
- 4 Complete the table at the right by calculating for each species of fish the total number caught and their combined mass.

Fisher	Type of fish	Mass of fish (kg)
Anastasia	Marlin	217
	Marlin	272
	Sailfish	68
	Mackerel	8
	Sailfish	75
Sally	Yellow fin tuna	33
	Mackerel	6
	Mackerel	11
	Sailfish	52
Brad	Mackerel	10
	Mackerel	7
	Mackerel	7
	Yellow fin tuna	41
	Marlin	377
	Mackerel	9
Omar	Sailfish	95
	Mackerel	7
	Mackerel	4

Species of fish	Total no. caught	Total mass caught (kg)
Marlin		
Sailfish		
Yellow fin tuna		
Mackerel		

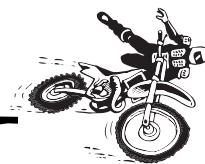
Tag and Release Programmes

Game fishing might bring on an adrenalin rush, but there's a serious side too.

The protection of marine resources is important. For many species of fish caught in Australian waters, there are regulations about how many fish can be taken in one catch, and the minimum mass and/or length of fish that can be legally caught and kept. With some species, it is illegal to keep any fish caught. These regulations aim to ensure that different species do not become scarce.

In Australia, tag and release programmes are encouraged by fishing associations, government departments and environmental groups. They involve the insertion of a tag into a caught fish, and the release of the fish back into the water. If tagged fish are re-caught, details such as location caught and length and mass of the fish are recorded.

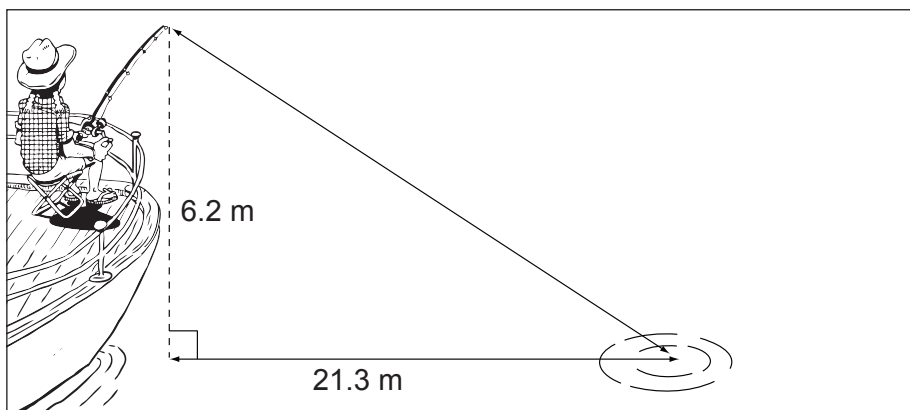
This data is sent to the agencies that run tag and release programmes, and is vitally important in the management of marine resources.



- 5** Using the completed table from question 4, construct two pie charts.
- In the first one, show a comparison between the number of fish caught, by species.
 - In the second one, show a comparison between the total mass of the fish caught, by species.

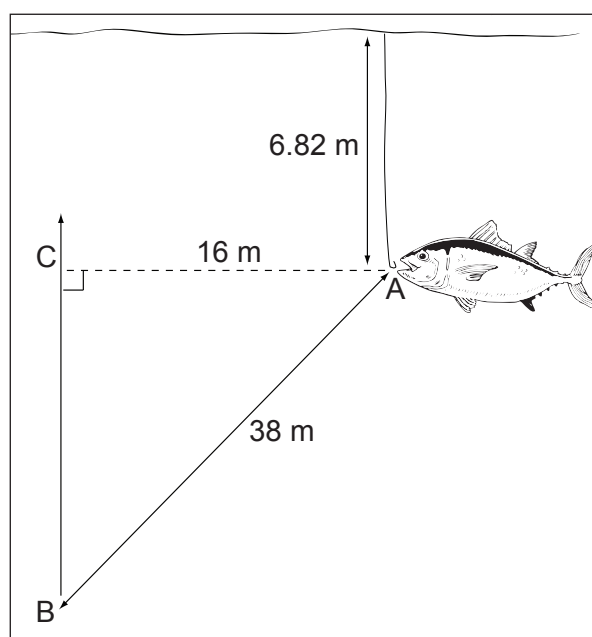
Tip: Don't forget the key!

- 6** As shown in the diagram, the tip of a fishing rod is 6.2 m above the surface of the water. The line, which is at 90° to the rod, enters the water a horizontal distance of 21.3 m from the tip of the rod.



What is the length of the fishing line from the tip of the rod to the point where it enters the water?

- 7** The person with the fishing rod in Question 6 is about to catch a Yellow fin tuna. As you can see in the diagram at the right, the Yellow fin tuna takes the bait 6.82 m below the surface of the water. It then dives along the line AB for a total distance of 38 m.



- If the horizontal distance AC is 16 m, what is the total depth reached by the fish at point B?
 - Suppose the tuna takes another 34.5 m of fishing line in its dive along the line AB. What is the total length of the line from the tip of the rod to the lowest point in the dive?
- 8** If the Yellow fin tuna from question 7 took 4.8 seconds to dive the 38 m from A to B after taking the bait, what was its average speed (m/s) during the dive?
- 9** As the Yellow fin tuna was reeled in, it hit the hull of the boat. If the tuna had a mass of 41 kg and was travelling at 2.9 m/s when it struck the boat, calculate its momentum when the collision occurred.

Hint: Momentum is a measure of a body's mass multiplied by its velocity (speed). Its unit of measurement is kg.m/s.

As you're calculating the tuna's momentum, here's something to think about – the momentum of that fish travelling at that speed is roughly equivalent to the momentum of a loaded shopping trolley moving at 30 km/h. The moral of that story is that in either case – you wouldn't want to be in its way!



Into your downhill skiing? Take it one step further and you're into extreme skiing territory! In Canada, skiers or snowboarders get to the country's more remote ski runs in a helicopter, which drops skiers off at the top of a slope and picks them up at the bottom.

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

One helicopter company offers extreme skiers the choice of skiing three, four or seven runs in a package. The longest run, Hellsgate, is 1 000 m in a straight line from top to bottom.

Three skiers take a helicopter ride to the top of Hellsgate. Jasmine took 5 min 38 sec to ski down the slope. Eli took 4 min 41 sec to ski down the slope, and Nik, took 6 min 52 sec to ski down the slope.

1 Express each skier's time as a decimal.

- a Jasmine: 5 min 38 sec = 5.____ min
- b Eli: 4 min 41 sec = 4.____ min
- c Nik: 6 min 52 sec = 6.____ min

2 Express each skier's time in hours. Answer to 3 decimal places.

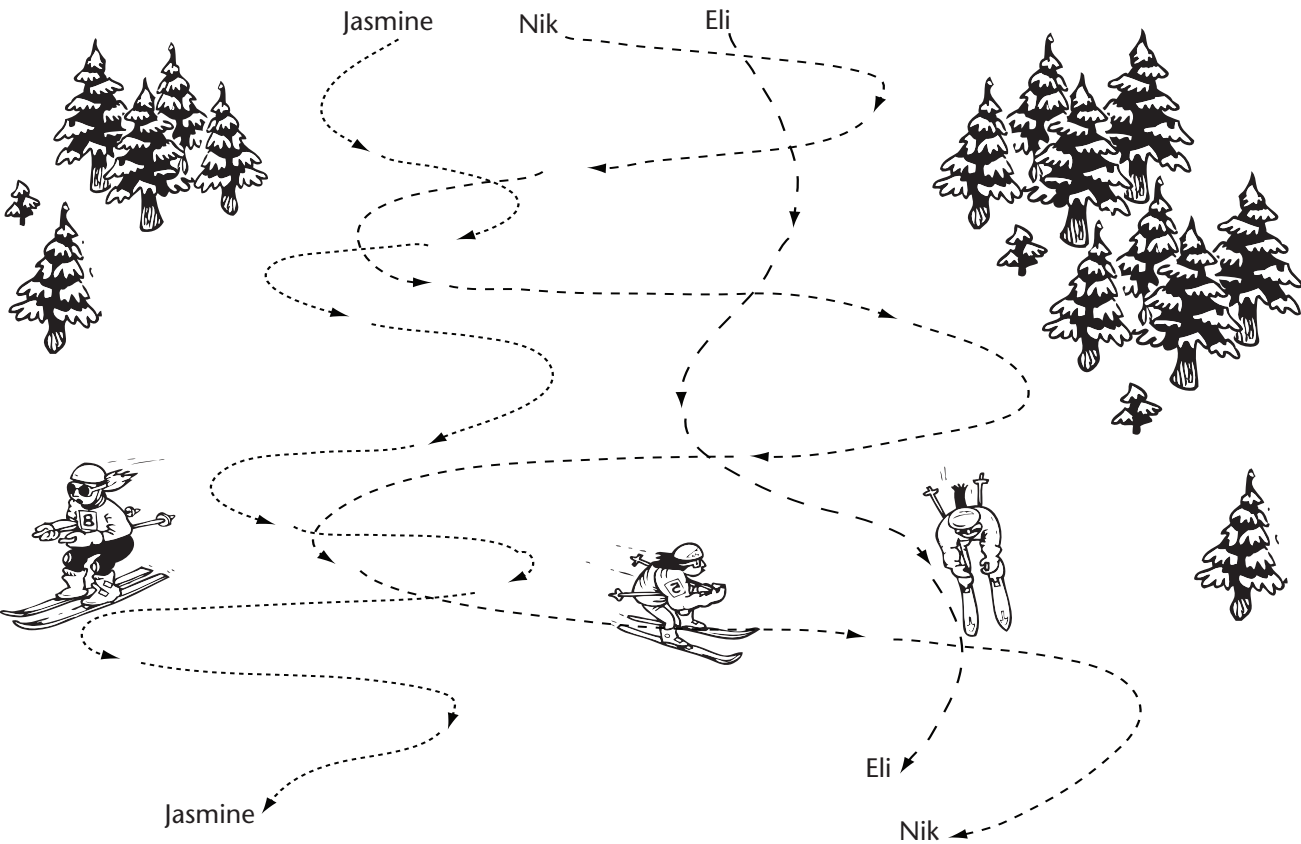
- a Jasmine
- b Eli
- c Nik

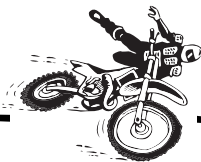
3 Imagine that each skier had skied in a straight line down Hellsgate from top to bottom – that is, a distance of 1 km – in the times given above. Complete the table by:

- a calculating each skier's average speed in km/h.
- b ranking the skiers in order from fastest to slowest average speed.

Skier	Average speed (km/h)	Ranking (fastest to slowest)
Jasmine		
Eli		
Nik		

The diagram below shows the route each skier actually took down Hellsgate.



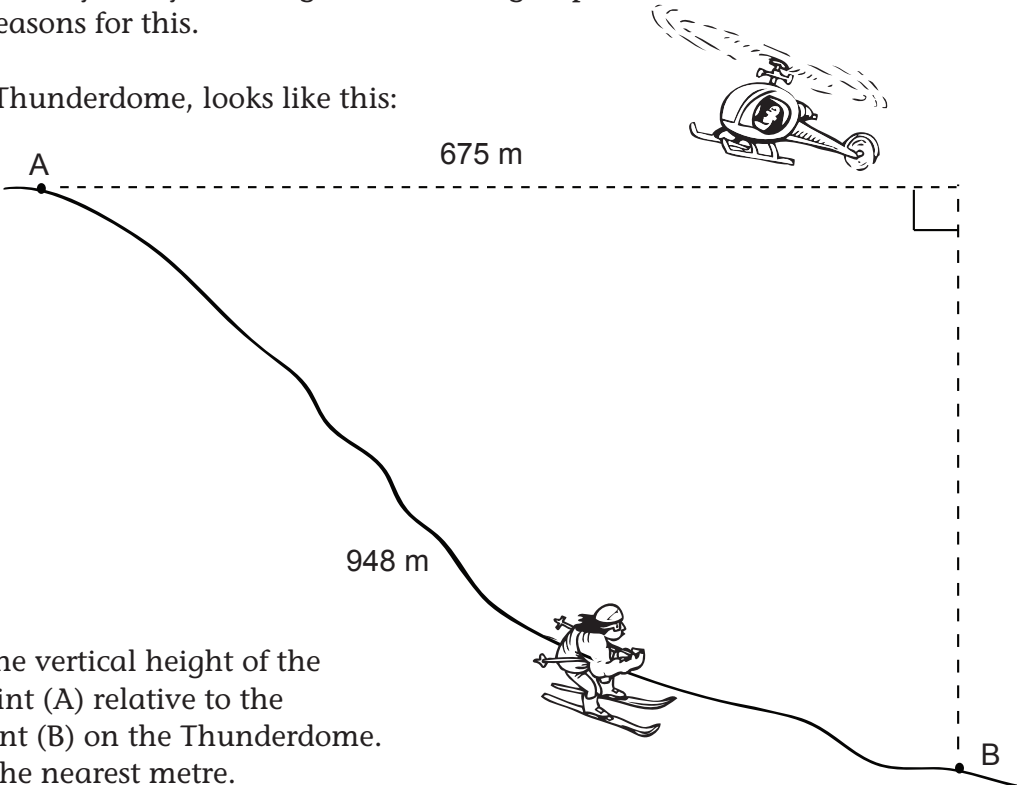


The actual distances covered by the skiers were:

- Jasmine 2 278 m
- Eli 1 544 m
- Nik 2 891 m.

- 4 Calculate the actual average speed of each skier.
- a Jasmine ____ km/h b Eli ____ km/h c Nik ____ km/h
- 5 a Now rank the skiers in order from fastest to slowest, in terms of their actual average speeds.
- b Is the order the same as that in question 3? Explain why this is the case.
- 6 In reality, skiers very rarely ski straight down a long slope. Suggest two reasons for this.

Another run, the Thunderdome, looks like this:



- 7 a Calculate the vertical height of the drop-off point (A) relative to the pick-up point (B) on the Thunderdome. Answer to the nearest metre.
- b If point B is 3 287 m above sea level, calculate the height above sea level of point A.

- 8 Mount Kosciusko is Australia's highest mountain. Its elevation is 2 228 m above sea level.
- a Does point A on The Thunderdome have a higher or lower elevation than Mount Kosciusko?
- b What is the difference in metres?



- 9 The rates the helicopter company charge per person are shown in the table. If the exchange rate for Canadian currency to Australian currency is \$1 AUD = \$0.83 Can, what are these charges in Australian dollars?

Package	Rate (Canadian dollars)
3 runs	\$720 Can
4 runs	\$795 Can
7 runs	\$895 Can

Skydiving - falling fast



7

So what does it feel like to fly? Skydivers know – jumping out of an aeroplane with a parachute is an incredible experience. Imagine the feeling of total freedom as gravity pulls you towards the Earth, which – for a short while anyway – seems like a long way down as you plummet towards it at 200 km/h! That's called free-falling. When the parachute opens, the ride downwards becomes a lot more controlled and you get to take in views like you'll see nowhere else – from 1 000 metres or more above the Earth.

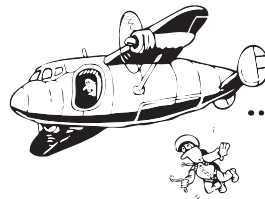
Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

Use the diagram to help you answer the following questions.

- 1** If the skydiver fell at an average speed of 200 km/h between Point A and Point B, how far had she fallen after 30 seconds?

Answer in kilometres. Then convert your answer to metres.

Hint: you need to convert 30 seconds to hours. When calculating the conversion, answer to 4 decimal places.



Altitude 2 700 m
Point A, Time = 0

- 2** How much further is there to dive between Point B and the ground?

Answer in metres. Then convert your answer to kilometres.



Point B
Parachute opens
Time = 30 s from
Point A

- 3** For this skydive, calculate the percentage of the total distance covered:

- a** between Point A and Point B.
- b** between Point B and the ground.



- 4** What is the total time the skydiver spent in the air?

- 5** For this skydive, calculate the percentage of the total time spent in the air:

- a** between Point A and Point B.
- b** between Point B and the ground.



Ground
Time = 5 min 20 s
from Point B

- 6** Complete the sentence:

In this dive, the skydiver spent just 8.57% of the time in the air covering _____% of the total distance. The remaining _____% of the time in the air was spent covering _____% of the distance.

Skydiving - falling fast (cont'd)



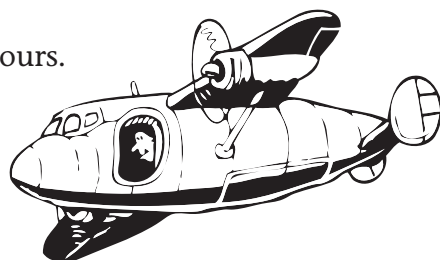
7

7 a What is the skydiver's average speed in m/s between Point B and the ground?

b What is this figure in km/h?

Hint: use 4 decimal places when converting seconds to hours.

8 How much slower in km/h is the skydiver's average speed between Point B and the ground, compared with the average speed between Point A and Point B?



9 Construct two line graphs.

- In the first graph, show the relationship between the distance fallen by this skydiver and the time spent in the air.
- In the second graph, show the relationship between the skydiver's average speed and the time spent in the air.
- In both graphs, time should go on the x -axis (horizontal axis).
- In the first graph, distance fallen should go on the y -axis (vertical axis). In the second graph, average speed should go on the y -axis.
- Remember, you will need to work out a suitable scale for your graphs.
- On both graphs, form the line by joining the three points that can be plotted from the information given. Before constructing the graphs, draw up two tables of x and y values, showing in each one the three sets of co-ordinates that can be plotted.



10 In skydiving competitions involving the accuracy with which divers can land on or close to a target, teams of five people compete over eight rounds, accumulating points as they go.

The winning team is the one with the least points on the day. Typically, the height from which they jump is 2 700 m. The 'dead centre' of the target is a disc 2 cm in diameter, and measurements are made in 1 cm intervals for the distance away from dead centre each skydiver lands, up to 16 cm away.

If you hit dead centre, the score is 0.00. If you land 16 cm away, the score is 0.16 (that is, 0.16 of a metre away). Scores are added and count both for individuals and for teams.

a A skydiver scores 0.09 for one jump.
How far away from dead centre did she land?

b The same skydiver has a total score of 0.33 after three jumps. What is her average score for each jump?

c Was the score for the first jump above or below her average score?
By how much?

d If the score for her third jump was equal to the average score for the three jumps, was the second jump scored higher or lower than the average score? By how much?

e What score did she receive for her second jump?



Skydiving - formations & handicaps



What's more extreme than going skydiving yourself, or with friends? Doing it while in the thick of fierce competition with other skydivers, that's what! It all unfolds thousands of metres up in the air ...

In competitive skydiving, points are awarded to individuals or teams for performance. In formation skydiving, teams gain points for performing patterned routines while free falling (dropping through the air without their parachutes open).

The two types of formations are known as **randoms** and **blocks**. Randoms involve the team making a single formation. Blocks are double formations, with a specific movement pattern between them.

For example, a block formation could involve four team members linked together in a circle, then each team member separates, does a 360° turn, and links up in the circle again.

Blocks are worth 1 point for each correct formation – that is, a maximum of 2 points – and randoms score 1 point.



Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

- 1
- In a skydiving competition, Round 1 consisted of two blocks and one random. The results for that round are shown in the table at the right.
- a

How many points in total was it possible to score in this round?
- b

List the teams and their scores in order from highest score to lowest score after Round 1.

Round 1			
Team	1st block (pts)	2nd block (pts)	Random (pts)
Blitz	2	1	–
Lightning	2	2	1
Dragonz	1	2	1

Here are the results for the next three rounds:

Round 2					Round 3				Round 4		
Team	1st block	1st random	2nd block	2nd random	1st random	2nd random	1st block	2nd block	1st block	2nd block	Random
Blitz	2 pts	1 pt	1 pt	1 pt	1 pt	1 pt	2 pts	2 pts	1 pt	2 pts	1 pt
Lightning	1 pt	1 pt	2 pts	1 pt	–	1 pt	2 pts	1 pt	1 pt	2 pts	–
Dragonz	2 pts	1 pt	2 pts	1 pt	1 pt	1 pt	2 pts	1 pt	2 pts	1 pt	1 pt

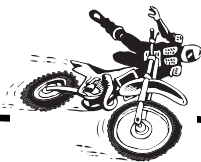
- 2
- a

How many points in total was it possible to score in Rounds 2, 3 and 4?
- b

List the teams and their scores in order from highest score to lowest score after **all four** rounds.
- 3
- In competitive skydiving, a handicap system is sometimes used when teams compete against one another.
- a

What is the purpose of using a handicap system?
- b

Identify one other sport where handicap systems are used.



The handicap system in competitive skydiving works on a formula which compares the average results achieved by various teams. Handicapping works by establishing a comparison between the top team in any one season with the other teams competing against it.

The top team prior to this competition was Dragonz, and so this team has no handicap. The other two teams both have a handicap, which is a number by which their points are multiplied before comparing final results in a competition.

- The handicap for the Blitz team is 1.06.
- The handicap for the Lightning team is 1.12.

- 4 a Calculate the scores after four rounds, taking into account the handicap for the Blitz team and the Lightning team.
- b Which team is leading after four rounds?

In this competition there are a further three rounds, to make a total of 7 rounds. The points scored in total for Rounds 5, 6 and 7, factoring in handicaps, are:

- Blitz – 14.84 pts
- Lightning – 16.8 pts
- Dragonz – 17 pts

- 5 How many points in total were scored in Rounds 5, 6 and 7 by each team **before** the handicap was factored in?

- 6 What were the final scores for this competition? Complete the table to find out who won on the day.

Team	Raw points (without handicap)	Handicap	Final score (with handicap)	Ranking on the day
Blitz		1.06		
Lightning		1.12		
Dragonz		–		

- 7 At the end of the skydiving season, the final competition scores (factoring in the handicap) were:

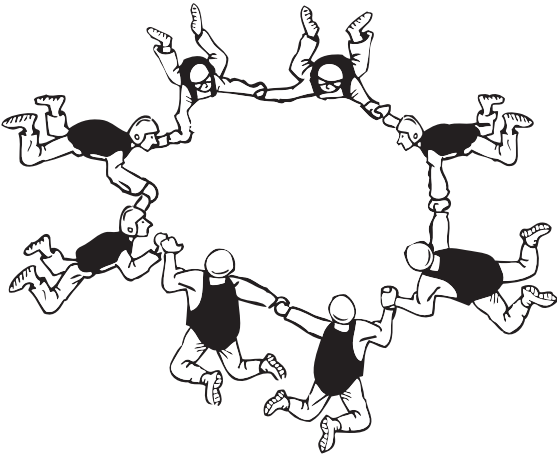
- Blitz 245.92 pts
- Lightning 274.4 pts
- Dragonz 221 pts.

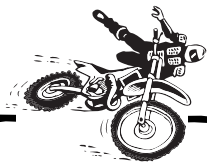
- a How do you think the handicap system worked for this season? Was it fair? Why?
- b Is looking at the scores at the end of a season enough to make a judgement on whether the handicap system is fair or not? What other reasons might explain why the team that was best in the previous season came last by a relatively long way?

- 8 Each skydiver in a formation team of 8 performs a block where the following moves occur.

- The eight skydivers form a circle and rotate as a group 180° in one direction and back through another 180° rotation to their original starting positions.
- They release grips entirely so each skydiver can do an individual 360° rotation.
- The second formation consists of all skydivers joining into a circle again and, as a group, performing a full 360° rotation.

Through how many degrees in total does each skydiver rotate in this block?





This is one extreme sport – for participants and spectators alike! V8 Jetsprint racing began in New Zealand in the 1980s, and is now also staged in Australia and the USA. It’s sometimes called rally driving on water – these boats aren’t that big, but they’re a whole lot more awesome than your average rally car. They complete 180 degree corners, pulling a similar magnitude of force to an F-111 fighter jet, and their acceleration is mind-boggling. Imagine putting a drag racing car’s engine into your average aluminium fishing runabout, and you’re close to the mark.

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

- 1 Jetsprinting boats can accelerate from 0 to 160 km/h in 2 seconds. Assuming the acceleration is constant over the 2 seconds, what would be the boat’s speed after:
a 0.5 s? b 1.0 s? c 1.5 s? d 2.0 s? e 1.34 s?
- 2 In one race in the USA, a Jetsprinting boat accelerated from 0 to 120 km/h in 4 seconds, then maintained this speed for a further 4 seconds. Assuming constant acceleration, complete this table.

Time (s)	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	5.0	6.0	7.0	8.0
Speed (km/h)	0								120				

- 3 Use the table in question 2 to construct a line graph showing the speed of the boat against time. Put time on the x-axis (horizontal axis) and speed on the y-axis (vertical axis).

Some Jetsprinting boats are powered by engines that produce up to 1 200 horsepower (hp). Horsepower is a unit of measurement of power used in relation to engines. Its use is not as common now as it once was. Today, the most common unit of measurement for power is Watts (W). When measuring engine power, 1 hp is approximately equal to 746W.

- 4 a What is the power output of a 1200 hp engine in Watts?
b Now convert your answer to kilowatts (kW).
- 5 Jetsprint boats have a typical mass of 500–600 kg. They have one of the highest power-to-weight ratios of any boat, car or other vehicle used in motorsports.



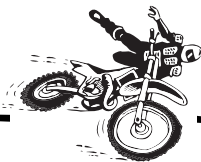
- a If a Jetsprint boat weighed 600 kg and was powered by an engine with a power output of 850 kilowatts (kW), what is the power-to-weight ratio of this vessel? (ratio of kW:kg)
- b Why do you think power-to-weight ratio is an important consideration in motorsports?

A Jetsprinting course was constructed over an area roughly equivalent to a football field. The channels are between 4 and 8 metres wide, and 1 metre deep. The diagram on the next page shows how a typical course might look.

In a race, there are three qualifying rounds where competitors are timed around the course. Each boat carries a crew of two – a driver and a navigator. The top 16 then race off in a series of ‘sudden death’ rounds, where only the top half of the competitors proceed to the next round. The field goes from 16 to 8 to 4 and then the final 2, who race for top spot. The boats are required to go around the course, passing the numbered points in a certain sequence.

A typical race sequence for the course in the diagram might be:

START, 1, 2, 9, 10, 5, 4, 6, 3, 8, 7, 10, 1, 8, 2, 9, 8, 3, 7, 10, 5, FINISH

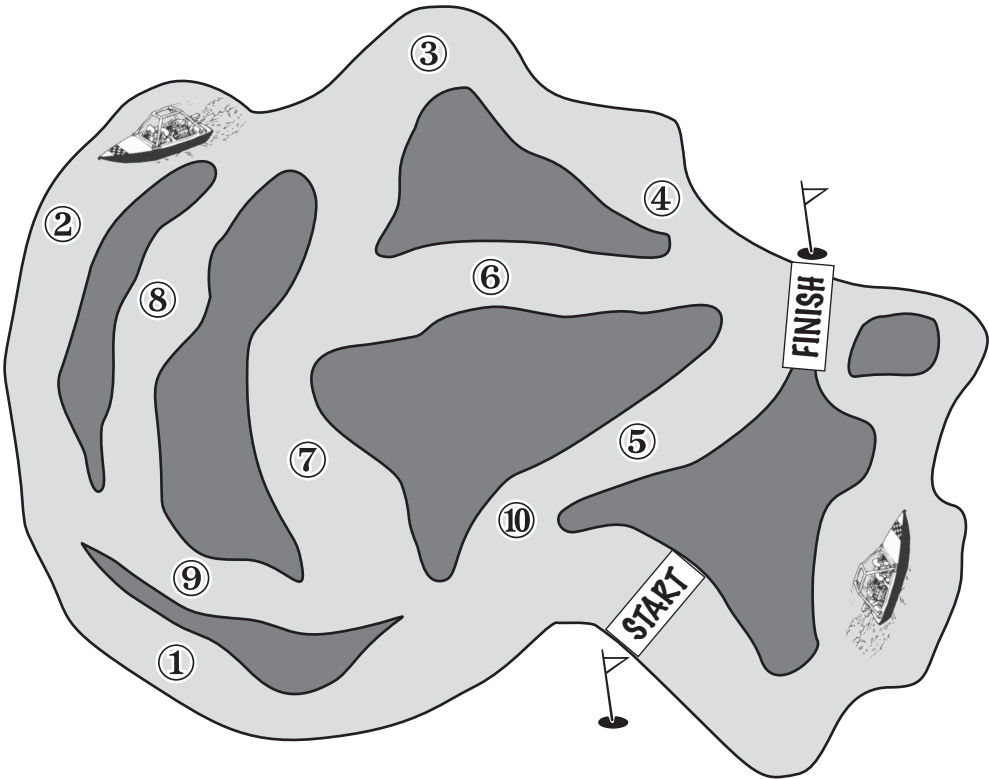


6 The fastest way for a Jetsprint boat to reverse its direction is to do a 180 degree turn, which its engine and the course design allow. Following the course sequence provided on the previous page and the diagram, identify the seven places within the sequence at which a driver would do a 180° turn, and which number they would head to next. The first one is done for you.

- 2 ➔ 9 – Travel from 1 to 2, then 180° turn to get back to 9.
- ____ ➔ ____
- ____ ➔ ____
- ____ ➔ ____
- ____ ➔ ____
- ____ ➔ ____
- ____ ➔ ____
- ____ ➔ ____

Here are the results of a ‘sudden death’ round of 16.

Team	Time (min, sec)
1	1 12.65
2	1 08.71
3	0 59.82
4	1 03.77
5	0 58.98
6	1 11.26
7	1 13.02
8	1 01.33
9	1 15.84
10	1 07.08
11	0 59.63
12	0 59.32
13	1 06.19
14	1 10.70
15	1 09.76
16	1 06.10



- 7 a To what fraction of a second are the times in the table at the left measured – tenths, hundredths or thousandths?
- b List in order from fastest to ‘slowest’ the teams that made it through to the next round of 8.
- 8 a From the table, what percentage of the teams completed the course in less than 1 minute?
- b What was the difference in times between the first qualifier in the round of 8 and the last qualifier?
- c By how much did the ninth-placed team miss out on qualifying for the round of 8?
- d Which team was clocked with the slowest time for the course?
- e How much slower was this time than that of the fastest qualifier for the round of 8?

- 9 In the round of 8 that followed this ‘sudden death’ round of 16, three teams completed the course in less than a minute.
- a What percentage of teams in the round of 8 turned in times of less than a minute?
- b Is this more or less than the percentage of teams with sub-one minute times in the round of 16? By how much?

It's mayhem when extreme sport meets extreme logic and problem solving – have a bash and see how you fare!

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

1 Women's surfing sees some amazing waves and amazing rides. Here, we've projected into the future and summarised the five biggest waves surfed in a competition in 2020.

Your job is to correctly fill out the table below using the following information:

Surfer	Beach	Wave height	Month
Jenny White			
Ingrid Vermullen			
Inez Zammut			
Lisa Telford			
Hayley McIntyre			

In random order:

- the rides occurred in: Hawaii (Oahu); Johanna in Australia; Tamarindo in Costa Rica; Tavarua in Fiji; and Jeffreys Bay in South Africa
- the wave heights were 5.4 m, 6.8 m, 5.2 m, 6.5 m and 5.7 m
- competitions during the year took place in August, July, March, October and April.



- Vermullen surfed the smallest wave.
 - The biggest wave was ridden later in the year than any others.
 - White's ride was on a wave that was 0.3 m smaller than the one recorded in August, but bigger than the wave recorded in July.
 - McIntyre's ride and the one in South Africa occurred seven months apart.
- Telford is Australian and White is from Hawaii, but they recorded their rides at beaches in each other's country of birth.
 - Zammut's impressive ride on the second biggest wave for the year was in the Southern Hemisphere.
 - This year was the first time in three years that the biggest wave was not recorded in Costa Rica.

2 Four teams competed in a World Cup group for a downhill skiing competition.

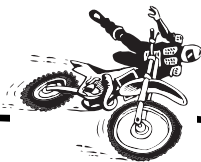
Various events are staged and scored, but penalties are also picked up during competition. In the end, the penalties are subtracted from the points gained to give a total score. It is the total score that determines the rankings.

Complete the table below, using the following information.





Team	Rank	Points	Penalties	Total
1				
2				
3				
4				



To establish points:

- The third ranked country:
Points = $4\,082.4 \div 3.6$
- Japan: Points =
 $(5\,768 \div 2) \times 3 - 7\,004$
- Hungary: Points =
 $2\,150 - (11 \times 25)$
- The fourth ranked country:
Points = $\sqrt{712\,336} + (95.5 \times 4)$

To establish penalties and totals:

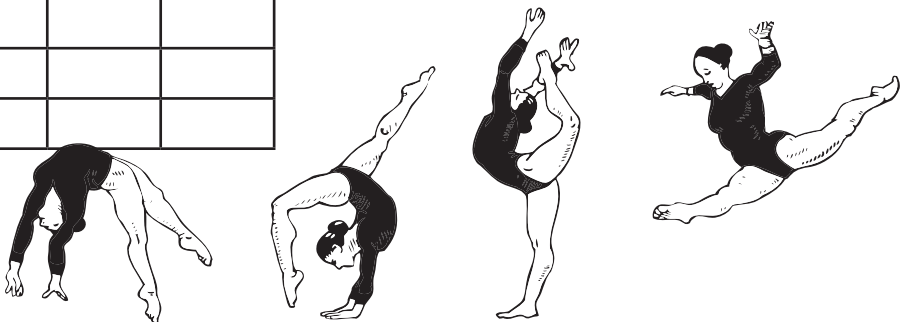
- Norway scored the highest number of penalties.
- When you multiply the first ranked team's points by its penalties, the answer is 311 250.
- The second ranked team scored 39 fewer penalties than the first ranked team.
- The third ranked team scored the fewest penalties to give it a total score of 1 028.
- There were only 18 points in total between Norway and Chile.

3 Elite level international gymnastics is about as extreme as it gets – finely tuned athletes perform adrenalin-pumping routines, based on strength, balance and precision timing. In the recent ‘Hypothetical World Championships’ held in Beijing, four of the medal events for women were the balance beam, uneven bars, vault and floor. Each event was held on a different day – Saturday through to Tuesday – and each event was won by a competitor from a different country – Japan, China, Korea and the USA. Four judges awarded their highest scores for the competition in these four medal events. Your job is to complete the table below using the information given, to show which country won which event on which day, and in which of the events each judge awarded their highest scores.

Suggestion: In this table, when you work out a correct event, day or nationality of judge, place a ✓ in the cell in the table. Then place a X in each of the other cells to which that selection cannot apply.

	Japan	China	Korea	USA
Balance beam				
Uneven bars				
Vault				
Floor				
Saturday				
Sunday				
Monday				
Tuesday				
Canadian judge				
Austrian judge				
Chinese judge				
Russian judge				

- The Chinese judge awarded her highest score on Sunday, but it did not go to the competitor from her own country.
- The Balance Beam was the only event where no Asian competitors finished among the medals.
- Korea was expected to take gold in the Uneven Bars on Saturday, but in the end, no one could beat the host country.
- Neither the Austrian judge, whose highest score was awarded for the Floor event, nor the Canadian judge gave their highest score on the weekend.
- The USA’s gold medal came on Monday, two days after the Uneven Bars event.
- The vault event was not held on Monday. Korea did not qualify for the final in this event.





Nothing pumps the adrenalin like white-water rafting. Inflatable boats tumble and belt their way through rushing, gushing, swirling rapids – while the rafters inside use their paddles, reflexes, judgement and sometimes sheer brute strength to stay on course.

People of any level of experience and expertise can enjoy white-water rafting, although to successfully navigate some of Australia’s more rugged stretches of river, rafters should be both skilled and experienced.

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

The International White-water River Grading System

To assist rafters in determining the degree of difficulty likely to be encountered when paddling a white-water river, the International White-water River Grading System has been established. This system grades rivers into one of six categories, as you can see in the box at the right.

- 1 Why is a grading system needed for white-water rivers?
- 2 Suggest why an international system of grading white-water rivers is preferable to, for example, an Australian system.

A white-water rafting tour company offers a choice of tours on different rivers.

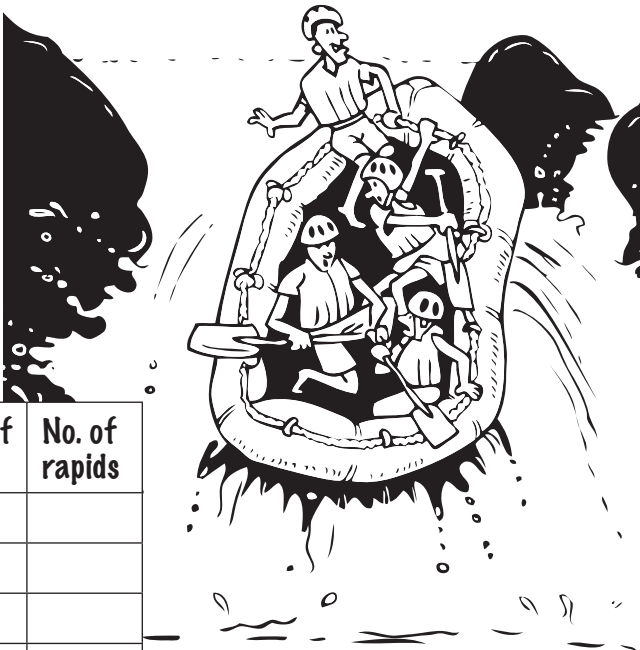
Tours available!

- Blue River: Grades 1, 2, & 3 rapids
- Black River: Grade 3 rapids
- White River: Grades 3, 4 & 5 rapids
- Brown River: Grades 4 & 5 rapids
- Grey River: Grades 2 & 3 rapids
- Red River: Grades 1, 2, 3, 4 & 5 rapids

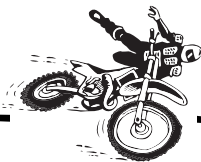
- 3 Which river(s) of those listed above would be most suitable for a non-adventurous beginner?
- 4 Which river(s) could be recommended for experienced rafters?
- 5 Complete the table at the right to show the number of each grade of rapids that can be experienced in the tours run by this company.

Grade of rapids	No. of rapids
1	
2	
3	
4	
5	

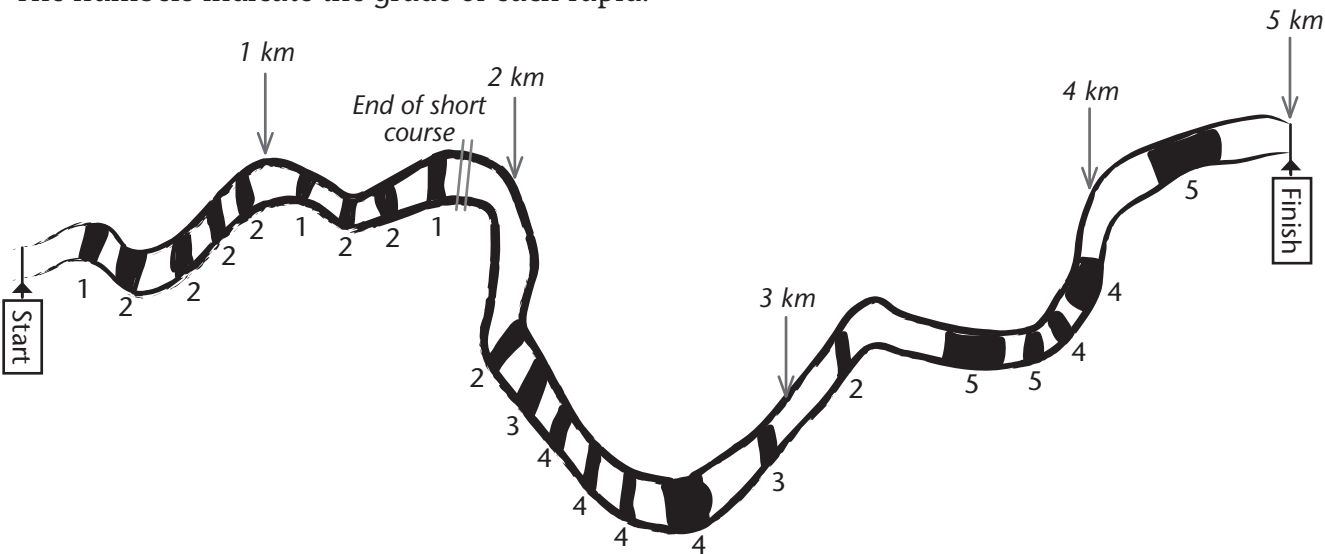
- Grade 1 – rapids consist of small, regular waves.
- Grade 2 – rapids have regular, medium-sized waves and some tactical moves may be needed.
- Grade 3 – waves of reasonable size; rafters likely to need to hold on at times. Technical manoeuvring of raft required to negotiate rapids.
- Grade 4 – difficult rapids with high powered and irregular waves. Continual manoeuvring of raft required to negotiate rapids.
- Grade 5 – very difficult rapids with large drops and fast currents. Complex and powerful manoeuvring required to negotiate rapids.
- Grade 6 – very dangerous rapids; extreme technical skill needed to negotiate rapids – not attempted by the vast majority of white-water rafters.



White-water rafting - the rapids (cont'd)



Look at this map of the Red River course. The black shading indicates rapids. The numbers indicate the grade of each rapid.



- 6 If a rafting crew completes the 1.8 km short course in 1 hour and 20 mins, what is their average speed in km/h along this course?
- 7 How many Grade 1 and Grade 2 rapids are passed on the Red River short course?
- 8 If the same rafting crew completes the entire course in 6 hours and 10 minutes, which includes 45 minutes for lunch, what is the crew's average speed while on the water?
- 9 a Was the average speed over the short course faster or slower than the average speed over the entire course?
b By how much? Answer in km/h.
c Convert your answer to part b to metres per hour.
- 10 Calculate the average speed of the rafting crew over the second part of the course – that is, from the end of the short course to the finish of the entire course.
- 11 Explain why the average speed for the second part of the course was different from the average speed over the short course.
- 12 In completing the entire Red River course, how many sets of Grades 1, 2, 3, 4 & 5 rapids are passed? Complete the table at the right.
- 13 a In which stretch of the course – that is, between which kilometre markings – is the highest number of rapids?
b How many sets of rapids are there in this stretch?



Grade of rapids	No. of rapids
1	
2	
3	
4	
5	

White-water rafting - the costs



12

As with any extreme sport, appropriate equipment is essential for white-water rafting. Fans of the sport usually need to travel some distance to reach the waterways to pursue their sport. One of the realities of pursuing most sports or hobbies is their cost. Just as Roula found ...

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

Roula decided to take up white-water rafting. She visited a supplier and priced the basic equipment she'd need:

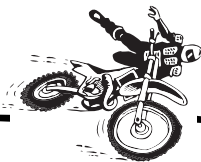
- Life jacket – \$230
- Paddle – \$140
- Helmet – \$90
- Raft – \$1 820
- Rope – \$40



- 1 How much in total would the equipment cost Roula?
- 2 Correct to 1 decimal place, what percentage of the total bill is represented by:
 - a the raft?
 - b the life jacket?
 - c the helmet?
- 3 Roula found out that if she purchased all five pieces of equipment online, she would receive a 15 per cent discount.
 - a How much would she save off the total price by purchasing online?
 - b If she purchased online, how much in total would she spend?
- 4 Roula also found that she could hire a raft at a cost of \$125 for a weekend. All the other equipment she needs could be hired for a further \$80 per weekend. She planned to go rafting on 12 weekends in a year.

Over the course of a year, would it be more or less expensive to hire the equipment rather than purchase it?
- 5 The supplier told Roula that if she purchased the raft for \$1 820 and it was in good condition after one year, they would buy it back from her for 45 per cent of the purchase price and use it as a hire raft.
 - a If Roula did this, how much would the supplier pay her for the raft after 12 months?
 - b How much would the raft have cost her for the 12 months if she sold it back to the supplier?
 - c How many times would the supplier have to hire the raft out before they had recovered the money paid to Roula for the used raft?
- 6 When looking at different rafts, Roula found that the dimensions of different models varied. One raft she saw measured 4.2 m long and 2.1 m wide. Another one was 5.5 m long and 2.4 m wide. These are the measurements of deflated rafts.

What is the total area occupied by each type of raft when deflated?



- 7 There are two white-water rafting locations that Roula is keen to visit. One is close enough to her home to visit in a day, but the other is further and requires a weekend.
- Roula calculated that the cost of fuel and provisions for the day trip would be \$160 each time, and the cost for fuel and provisions for a weekend trip would be \$280.
- If she purchased all her equipment at the online price calculated in question 3, and over the course of a year visited each of the two locations six times, how much would her rafting cost her for one year?
- 8 Using your answer to question 7, calculate Roula’s costs per rafting trip over the course of one year.
- 9 Over the summer holidays, Roula and three friends – Bronwyn, Callum and Ahmed – decide to go interstate on a rafting holiday.
- Roula has her raft and other equipment, but they need to hire another raft at a cost of \$330 for 10 days and three sets of equipment for \$120 each for 10 nights.
 - They calculate that fuel and other vehicle expenses for the holiday will cost \$600, and they also need to hire a trailer to carry their equipment at a cost of \$200.
 - Entry to the National Park they plan to visit is \$50, and camping fees are \$12 per person per night.
 - They have budgeted \$720 between them for food and other provisions for the holiday.

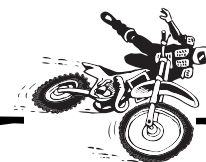
What is the total cost for the four of them to take the holiday?

- 10 The hire fees for the raft and equipment are being shared by Roula’s three friends. All other costs are being divided equally between the four of them.
- Complete the table to work out how much it costs each person for the holiday.



Cost	Total \$	Roula’s share	Bronwyn’s share	Callum’s share	Ahmed’s share
Raft hire					
Equipment hire					
Fuel					
Trailer hire					
National Park entry					
Camping fees					
Provisions					
TOTAL					

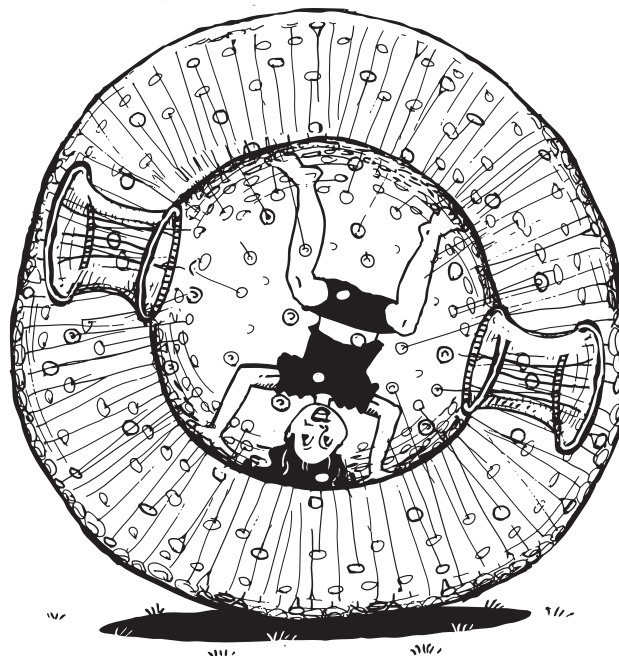
- 11 Using the first two columns of the table above, construct a pictogram to show how much money was spent in each category listed in the left hand column. The pictogram needs to show only the **total** money spent on each category, not how much each person spent.
- Use different symbols for each category, and choose symbols that each relate in some way to the category it applies to.



And now for something completely different! Sphereing or Zorbing (also sometimes called Globe-riding) involves participants travelling inside a sphere which is usually constructed of transparent, flexible plastic.

There is one sphere inside another sphere, with an air layer in between to act as a shock absorber. The inner and outer spheres are often joined by dozens of ropes. A person enters the inner sphere through a porthole.

The name 'zorbing' originates with three New Zealand business partners. They conceived the idea for a sphere, called it the 'zorb', and patented it and trademarked the name in 1994. They created the company Zorb Ltd, and set about franchising the business model.



Sphereing is generally done just for fun, not competition. The sphere usually moves down a gentle slope so the rider rolls downhill, but it is also possible to travel on a level surface or on water.

Most spheres are constructed for a single rider, but some hold two or three. The length of sphereing runs can be up to 0.81 km (0.5 mi), and they typically reach speeds of 32–48 km/h (20–30 mph).

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

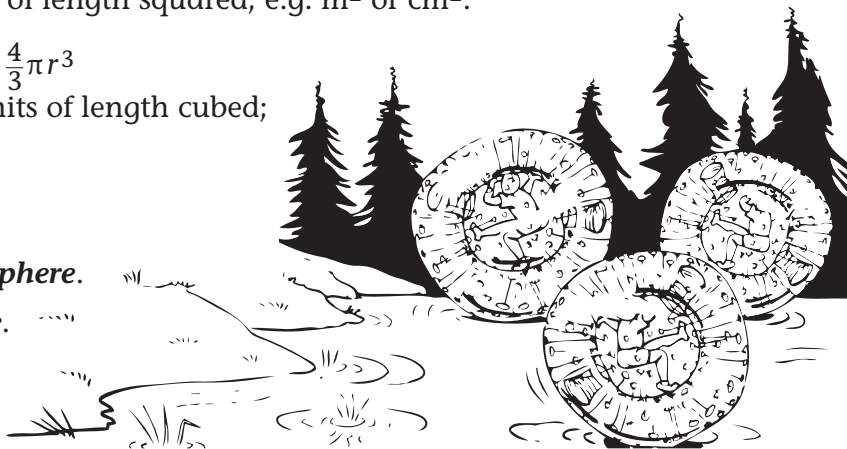
- 1 In 2006, the Guinness Book of Records officially recognised two world sphereing records – the longest continuous distance travelled, and the fastest speed.
Both records were made at the Matarua Farm near Rotorua in New Zealand. Steve Camp rolled the longest continuous distance of 570 m, and Keith Kolvers set a world zorb speed record of 52 km/h (32 mph).
 - a In the 2008 Beijing Olympics, Jamaican sprint champion Usain Bolt set a new world record for the 100 m sprint. He ran it in 9.69 seconds. Which speed was faster – Bolt's run at Beijing, or Kolvers' speed inside the sphere?
 - b The longest recorded golf drive in the world was hit in 1974 by Mike Austin at the National Seniors Open in 1974. Assisted by a 30 mph wind, Austin drove the tee shot a massive 1 545 feet. Was this a longer or shorter distance than Steve Camp's continuous roll in a sphere? By how much?
- 2 Some spheres contain a harness, which holds the participant in a relatively fixed position inside. Others allow participants to move freely within the sphere. If a person was strapped into a harness inside a sphere which rolled 100 m, and the sphere completed one complete rotation every 12.98 m, through how many degrees would the person rotate in that 100 m roll?
- 3 Spheres typically have a diameter of around 3 m.
 - a Define **diameter**.
 - b If a sphere has a diameter of 3 m, what is its radius?



A number of different quantities of circles and spheres can be calculated using formulae. These include the following.

- The circumference of a circle = $2\pi r$. Assume $\pi = 3.14$.
- The surface area of a sphere = $4\pi r^2$
Area is measured in units of length squared; e.g. m^2 or cm^2 .
- The volume of a sphere = $\frac{4}{3}\pi r^3$
Volume is measured in units of length cubed;
e.g. m^3 or cm^3 .

- 4**
- Define *circumference*.
 - Define *surface area of a sphere*.
 - Define *volume of a sphere*.



- 5** On a sphereing run in Queensland, the spheres in use have an outer diameter of 2.91 m.

- What is the radius of the outer spheres?
- What is the circumference of the outer spheres?
- If the circumference of the inner sphere is 6.15 m, what is the diameter of the inner sphere?
- If a person of height 173 cm was standing inside the inner sphere, how much space would there be between the top of the person's head and the surface of the sphere?

- 6** Calculate the surface area:

- for the outer sphere described in question 5.
- for the inner sphere described in question 5.

Remember, the units are squared!

- 7** Using the dimensions given for the spheres in question 5, calculate the volume of air held between the outer sphere and the inner sphere. Remember, the units are cubed!

- To do this, you will need to complete a three-stage calculation. First list the three stages.
- Now, perform those calculations!

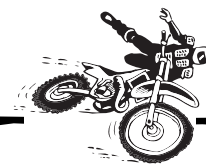
- 8** In question 5, you calculated the radius of the outer sphere and the diameter of the inner sphere.

- What is the radius of the inner sphere?
- Calculate the total volume of the outer sphere.
- Calculate the volume of the inner sphere.
- Calculate the volume of air held between the outer and inner spheres.



- 9** A standard men's basketball has a diameter of 23.8 cm.

- Calculate the volume of this basketball.
- Using your answer to question 8b, how many basketballs would fit inside a sphere with the dimensions given in question 5?



Skateboarding – pastime, means of transportation, and extreme sport! Today, mind-blowing speeds and tricks on skateboards have reached levels that were not even dreamed of just a couple of decades ago. As with any extreme sport, there are always those who want to push the boundaries that little bit further ...

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

- 1** Danny Way from California, a renowned extreme skater, holds a number of records. In 2004, he set the world record for the longest jump and the highest air off a 'Mega Ramp' he designed. The jump measured an impressive 79 feet in length, and 23.5 feet in height. The same style of 'Mega Ramp' was used in Way's successful jump over the Great Wall of China in 2005, where he became the first human to jump the wall without the aid of a motor.

Imperial units of length

12 inches (in) = 1 foot (ft)
3 feet (ft) = 1 yard (yd)
5 280 ft = 1 mile (mi)
1 760 yd = 1 mi

- a** How long and high were Danny Way's jump records, in metres?
b Danny reached an amazing speed of 55 mph before leaving the ramp. Read 'Did you know?' box about Joel King's record on a jet-powered skateboard.

What percentage of Joel King's speed did Danny Way achieve without a motor?

- c** If speed on leaving the ramp is the only factor in determining the height and length of a skateboard jump, what height and length of jump would be achieved by exiting the ramp at the same speed Joel King travelled on his jet-powered board? Answer in feet, and then convert your answer to metres.
d Speed on exiting the ramp is clearly not the only factor that determines how long and high a jump is. What other factors do you think might influence the jump?

Did you know?

On 29 August 2007, Englishman Joel King clocked up an astounding 110 mph (177.1 km/h) on a jet-powered skateboard ... Now that's extreme! He laid down on his back on the 1.5 m long board, which had a kerosene engine strapped to the back. It took him just over 2 seconds to reach the record speed. The board, which was constructed without any steering mechanism or brakes and cost King £7 000 (about \$16 000) to build.

American Gary Hardwick clocked 63 miles per hour – about 101 km/h – on a skateboard without any help from engines.

Biker Sherlock holds the record for the fastest speed while being towed by a motorcycle – 90 mph, or 145 km/h ... Insane!

- 2** Chad Fernandez, an American professional skateboarder, performed the longest skateboard rail grind – 10.31 m (36 ft 9 in or 36.75 ft). If the rail length was 15.00 m in total, what proportion of the rail was covered by Fernandez's grind?

- 3** Using your answer to question 2:
a what is this figure as a fraction in its simplest form?
b what is this figure as a percentage?

- 4** Another skateboarder competing in the grind-off with Fernandez achieved a grind length of 8.99 m. How far short of Fernandez's record was this grind?

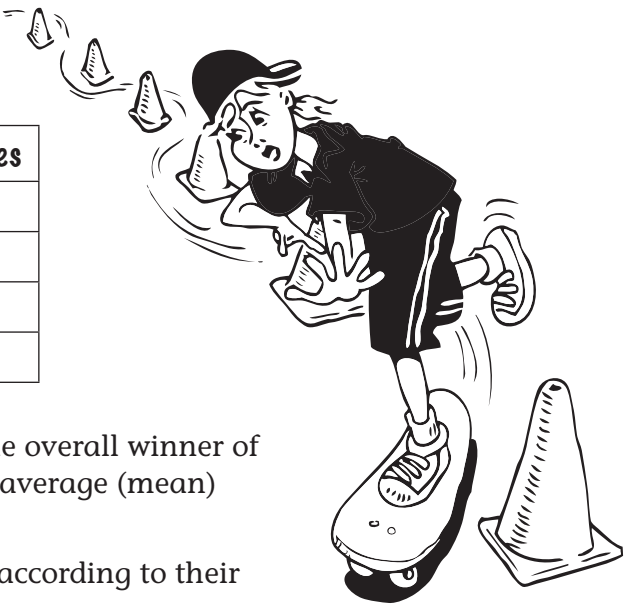
- 5** World rail-grinding records on skateboards are considerably shorter than those set on snowboards and skis. Identify one main difference between grinding on skateboards and skis/snowboards that accounts for this.



- 6 Slalom competition in skateboarding involves placing a course of cones along a road, and the skaters weave between them. Slalom courses vary in length, but typically consist of 40, 60 or even 100 cones – placed 2 m (about 6 ft) apart. Competitors are judged by the time they clock for the course, and how many cones are touched or knocked over on the way. The record for slalom skateboarding around 100 cones placed 6 ft apart was set by Martin Sweeney of the UK in 1991. Sweeney did the course in an incredible 21.7 s – without knocking over a single cone. The previous world record for the same slalom course was 25.27 s.
- a How much time did Martin Sweeney shave off the world record?
 - b What percentage did he take off the previous record?
 - c On average, how often did Martin Sweeney skate past a cone on his record-breaking slalom run?

7 The table below shows the results of a women's slalom skateboarding competition.

Competitor	40 cones	60 cones	100 cones
J. Nguyen	19.31 s	25.71 s	34.58 s
R. Sampson	16.99 s	22.16 s	31.64 s
W. Forsyth	14.75 s	20.04 s	33.98 s
Y. Ibrahim	18.23 s	27.88 s	36.16 s

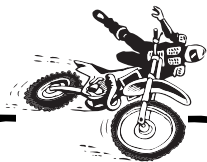


- a The women's scores are averaged to find the overall winner of the slalom event. Calculate each woman's average (mean) time for the three slalom courses.
- b Rank the competitors in order from 1 to 4, according to their average times across the three slalom courses.
- c Construct a column graph to show the results of this competition. Put the four competitors on the x -axis (horizontal axis). Put the times on the y -axis (vertical axis). Use colour coding to show the times for all three slalom courses (40, 60 and 100 cones) in a single column for each competitor. Make sure you include the key for the colour coding on your graph.

The Ollie

The ollie is a skateboarding trick invented by Alan 'Ollie' Gelfand in 1976, and this trick transformed the sport. It is an aerial manoeuvre which forms the basis of many other skateboarding tricks, such as the kick-flip, the heel-flip and the pop-shoveit. The ollie involves becoming airborne on a skateboard without holding the board. Danny Wainwright from England holds the official record for the highest ollie at 44.5 inches (113 cm).

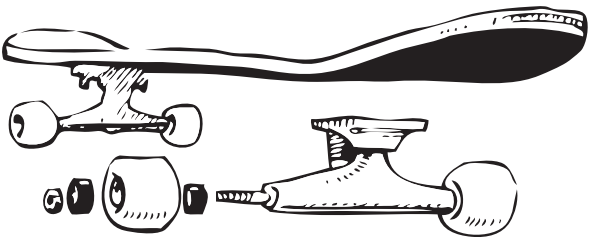
- 8 A skateboarder sets up a jump by building a temporary brick wall which is the equivalent of 10 house bricks high. She intends to practise her ollie technique by jumping over the bricks. The bricks have the dimensions 230 mm long $\times$ 110 mm wide $\times$ 76 mm high.
- a Sketch a house brick in a 3D drawing, showing the dimensions given.
 - b What is the total height of the jump in mm? Now convert this figure to cm.
 - c If the skateboarder successfully performs an ollie to jump the bricks, how far short of Danny Wainwright's world record is this jump? Answer in cm.



Whether it's at elite level or street level, skateboards take a fair bit of punishment. It's critical for a skater's equipment to be built to take what's dished out to it. Let's look at the bits and pieces that make up a skateboard, and some of the specifications that go along with them.

There are four main components to a skateboard:

- the deck
- the trucks (the axle that connects to the underneath of the deck)
- the wheels
- the bearings (inside the wheels).



So whether you're putting together your own skateboard from the components, or purchasing one complete – how do you decide what's right for you?

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

The Deck

A US skateboarding authority recommends the guidelines in the table below for skateboard decks. Measurements in the table on the right have been converted to centimetres.

Height of person (feet, inches)	Recommended length of deck (inches)
< 4 ft	< 29
4 ft–4 ft 10 in	29–30
4 ft 10 in–5 ft 3 in	30.5–31.5
5 ft 3 in–5 ft 8 in	31.5–32
5 ft 8 in–6 ft 1 in	32–32.5
> 6 ft 1 in	> 32.5

Height of person (cm)	Recommended length of deck (cm)
< 121.92	< 73.66
121.92–147.32	73.66–77.47
147.32–160.02	77.47–80.01
160.02–172.72	80.01–81.28
172.72–185.42	81.28–82.55
> 185.42	> 82.55

- 1

a

Use the figures in the tables above to write a formula for converting feet to centimetres.

b

Now write a formula for converting inches to centimetres.
- 2

Samantha is 121.92 cm tall and uses a skateboard deck that is 73.66 cm long. Her friend Adrian is 185.42 cm in height and uses a skateboard deck that is 82.55 cm long.

a

Calculate the ratio of height to deck length for both Samantha and Adrian.

b

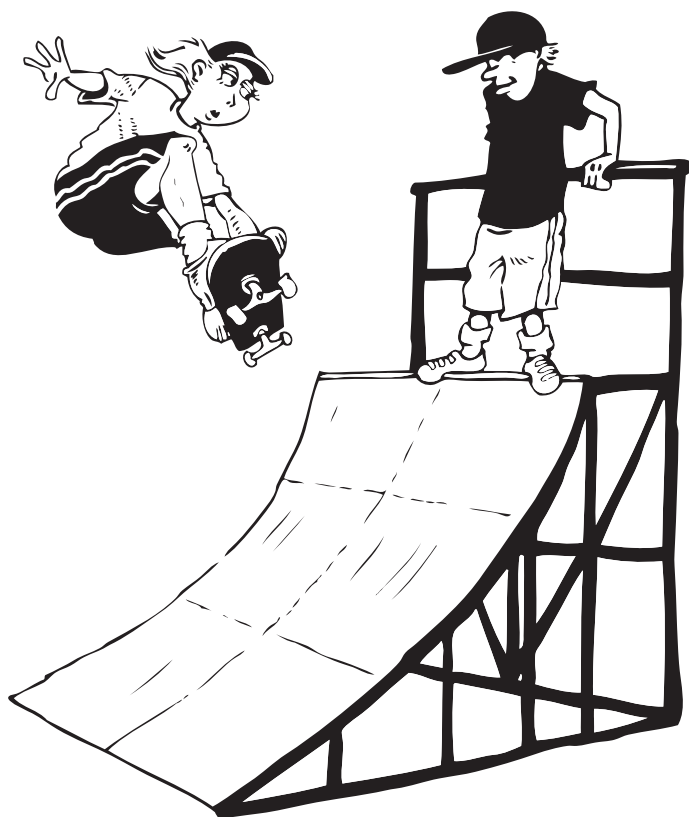
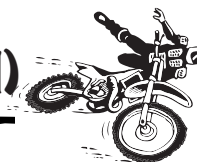
Whose height:length ratio is higher – Samantha's or Adrian's?
- 3

What is the height:length ratio for a skateboarder who is 172.72 cm tall with a skateboard deck length of 81.28 cm?
- 4

What statement can you make about the recommended ratio of a person's height to deck length, as the person's height increases?
- 5

Suggest why the recommended ratio of height to deck length changes as it does.
- 6

Would skateboard wheels with a larger diameter roll faster or slower than wheels with a smaller diameter?



The width of skateboard decks does not vary a great deal. Most are 7.5–8 inches wide (19.05–20.32 cm). The size of the skateboarder's feet is also a consideration – people with larger feet should use wider skateboard decks.

The Trucks

There are a range of different trucks available of varying quality, width and design. Essentially, trucks need to match the width of a skateboard deck, and be strong and flexible enough to do what a skateboarder wants his or her board to do.

The Wheels

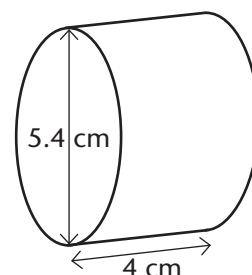
There are many different types of wheels. They are constructed differently – using different materials – and come in a range of diameters. The two main measurements to consider for skateboard wheels are wheel diameter and hardness.

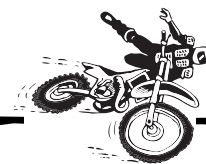
The hardness of wheels varies according to the types of surfaces they will be ridden on, and is given by a number. The higher the number, the harder the wheel. Typically, skateboard wheels have a diameter of 52–54 mm, with a hardness of 95–100a. Cruising skateboarders often prefer softer wheels, as they traverse rough terrain more often. Typical hardness of these wheels might be 75–85a.

- 7 Skateboarders who spend a lot of time riding ramps typically choose wheels of diameter 55–65 mm. Why do you think this is so?
- 8 A skateboard has wheels of diameter 56 mm.
 - a What is the circumference of the wheels? (Use the π value given by your calculator).
 - b What is the circumference of the wheel in cm? Answer to 2 decimal places.
- 9 By how much is the circumference of a skateboard wheel with a diameter of 65 mm longer than the circumference of a skateboard wheel with a diameter of 56 mm? Answer in cm.
- 10 The skateboard wheels in the illustration on the previous page are cylindrical in shape. Wider wheels have more surface in contact with the ground.
 - a What could be one advantage of having more wheel surface in contact with the ground?
 - b What could be one disadvantage of having more wheel surface in contact with the ground?
- 11 The surface area of an open-ended cylinder is calculated using the formula:

$$SA = 2\pi r \times \text{height } (h)$$

In the case of a cylindrical skateboard wheel, height can be taken as the width of the wheel. A wheel has a diameter of 5.4 cm and a height of 4 cm, as shown in the diagram. Using the formula above, calculate the surface area of the wheel. Your answer should be in cm^2 .





Luge is a winter sport where competitors lie on their backs on an amazingly fast, sled-like vehicle and fly down a twisting ice-covered course – typically between 1 000 and 1 500 km in length. They can reach speeds in excess of 130 km/h. Competitors are timed over a number of runs, and the person with the fastest combined times wins the competition. Luge competition is held in women's and men's singles and doubles and mixed doubles.

'Luge' is the French word for 'sled'. The luges used in elite competition, including the Winter Olympics, consist of a fibreglass shell with steel runners that slide over the surface of the ice. A luge can be steered, but it has no brakes. Once the luge is in motion, the competitor lies down and completes the course. To stop it, they sit up and pull the front of the luge upwards.

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

- 1 In a recent international luge competition in Finland, the length of the course for men was 1 463 m. The women's course was 917 m in length.
 - a What was the length of the men's course in kilometres?
 - b What was the length of the women's course in kilometres?
 - c How much longer than the women's course was the men's course? Answer in metres, then convert your answer to kilometres.

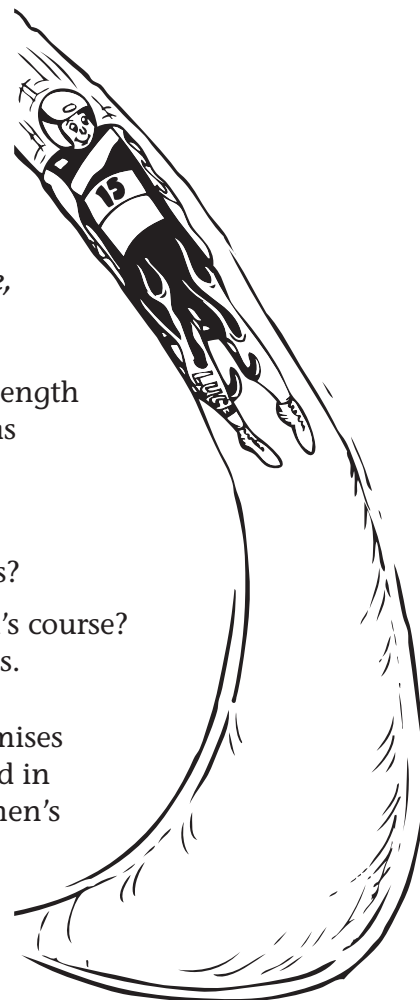
- 2 Luge courses are constructed to run downhill, which maximises possible speed during competitions. On the course described in question 1, the total vertical drop from the start of the women's course to the finish was 58.91 m. For the men's course, the vertical drop was 113.48 m.

On two separate graphs, show the gradient (slope) of each course. Put length of the course on the x-axis (horizontal axis) and vertical drop on the y-axis (vertical axis). Set up a suitable scale on the axes of each graph. On your vertical scale, make the height at the finish of the course 0 m.

- 3 From your graphs, which course had the higher gradient – that is, was steeper?
- 4 In international luge competition, the gradient (or grade) of courses is officially measured by expressing the ratio of the vertical drop to the length of the course as a percentage. To calculate this percentage, divide the vertical drop by the length of the course and multiply by 100. What is the grade of:
 - a the women's course?
 - b the men's course?

- 5 Use the information in question 4 to help you complete this sentence:

In international luge competition, the grade of the course is a measure of the _____ of the vertical drop to the _____ of the course. This figure is multiplied by 100, so it is expressed as a _____. The _____ the grade, the steeper the course.





- 6** In the competition in Finland, the fastest time in the women's singles event was clocked at 29.932 s. The fastest time in the men's singles event was 40.785 s.

- a** Calculate the average speed in m/s for the fastest woman and the fastest man.
- b** Convert your answers to part a to km/h.

- 7** There are strict rules regarding the mass of a luge used in international competition. The mass of a single luge cannot exceed 50.5 lb and a double luge cannot exceed 59.5 lb.

Given that 1 pound (lb) = 0.45 kg, convert the maximum mass of each luge to kilograms. Express your answers to the nearest whole kilogram.

- 8** In 2002, the Winter Olympics were held in Salt Lake City, Utah, USA. The 1 316 m long course constructed for luge cost US\$25 million to build. Its equipment included 24 cameras, 8 scoreboards and 49 timing points.

One camera was placed at the start line of the men's course and another at the finish line, and the remainder of the cameras were placed at equal intervals along the course. How far apart were the cameras?

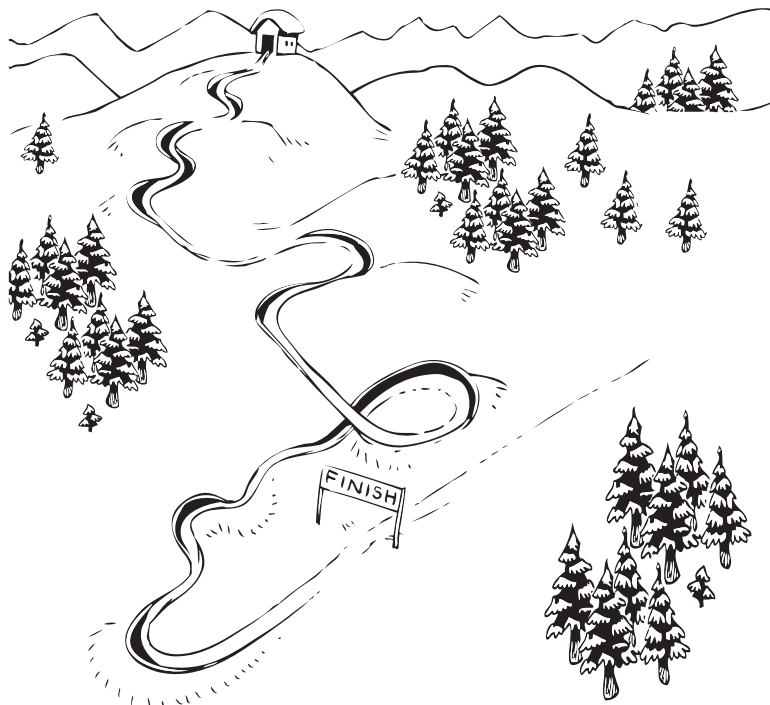
- 9** The Salt Lake City course contains a huge refrigeration system, with 90 km of pipes carrying 49 895 kg of ammonia refrigeration. It can maintain a temperature of -25°C . In a test cricket match at the MCG, the distance around the boundary line is 470 m. If the refrigeration pipes from the Salt Lake City luge course were placed around this boundary line, how many times would they circle it?

- 10** The mass of an average four-cylinder car is 1 200 kg. Do the necessary calculation, and complete this sentence:

To the nearest whole number, the mass of the ammonia refrigeration used at the Salt Lake City luge course is equivalent to _____ four-cylinder cars.

- 11** When athletes compete in the luge, they lie on their backs with their feet at the front of the luge and angled inwards.

- a** Suggest why this is the best position for riding a luge, rather than, for example, kneeling on it.
- b** What disadvantages are there with lying in this position while travelling 130 km/h down an ice track?



An insane record motorbike jump



17

Early on New Year's Day 2008, Australian daredevil motorbike rider, Robbie Maddison, set a new world record for a jump on a 500 cc motorcycle. Maddison jumped an amazing 322 ft in Las Vegas, smashing the previous record of 277 ft. In his record-breaking jump, he left the ramp at 94 mph, and his motorbike soared to a maximum height of 60 ft.

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

1 How far in metres was Maddison's jump?

2 By how much did Maddison beat the previous record:

- a in feet?
- b in metres?

Imperial units of length
12 inches (in) = 1 foot (ft)
3 feet (ft) = 1 yard (yd)
5280 ft = 1 mile (mi)
1760 yd = 1 mi

3 Complete the tables by correctly converting the units of measurement.

	feet	metres
Length of jump		
Maximum height		

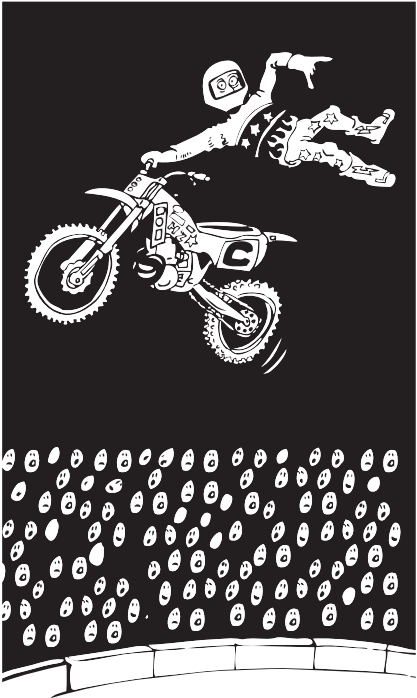
1 in = 2.54 cm
1 ft = 30.48 cm
1 ft = 0.31 m
1 mi = 1609.34 m
1 mi = 1.61 km
1 yd = 91.44 cm

	miles per hour	kilometres per hour
Ramp exit speed		

4 Suppose that during Maddison's record-breaking jump, the comparison between the distance covered and the height reached was given by the figures shown in the table at the right.

We've taken 2 ft off the total distance to make the calculations a bit simpler!

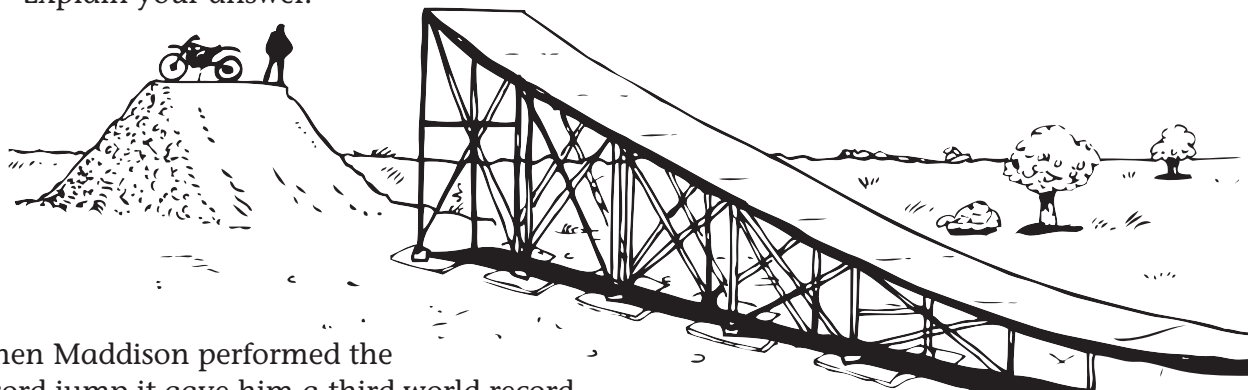
- a Complete the jump height and jump distance columns in metres.
- b While Maddison was ascending during his jump, what was the ratio of jump height to jump distance?
- c During his ascent, was this ratio constant? Explain your answer both mathematically and in terms of the feat performed.
- d Motorbike jumpers strive to achieve the right ratio between jump height and distance. Explain why this ratio is so important, and what could happen if a motorbike rider gained too much height too early into the jump.



Jump height (ft)	Jump height (m)	Jump distance (ft)	Jump distance (m)
0.0		0	
7.5		20	
15.0		40	
22.5		60	
30.0		80	
37.5		100	
45.0		120	
52.5		140	
60.0		160	
52.5		180	
45.0		200	
37.5		220	
30.0		240	
22.5		260	
15.0		280	
7.5		300	
0.0		320	



- 5 Using the table on the previous page, plot a graph showing jump height and jump distance in feet. Put distance in feet on the x-axis (horizontal axis) and height in feet on the y-axis (vertical axis).
- 6 If you had plotted the graph in question 5 using the height and distance figures in metres, how would the shape of the graph compare with the graph you plotted in feet? Explain your answer.



When Maddison performed the record jump it gave him a third world record.

At the time, he also held the record for the longest jump on a 250 cc motorcycle (221 ft) and the longest jump with a trick on a 500 cc motorcycle (246 ft).

- 7 Find out what cc stands for in relation to engine capacity, and exactly what is being measured to give the cc figures.
- 8
 - a What is the ratio of the larger capacity motorbike to the smaller capacity motorbike?
 - b What is the ratio of the lengths of the 500 cc jump to the 250 cc jump?
 - c Are the two ratios you calculated in parts a and b the same? Explain why.
- 9 We've looked at how motorbike riders aim for the right ratio between jump height and distance. Identify as many factors as you can that might affect the height:distance ratio achieved by a jumper on any particular occasion.

Another ratio that comes into consideration in stunt motorbike events is the power-to-weight ratio of the motorbike. The higher the power-to-weight ratio, the faster the bike will accelerate. A number of factors can contribute to the mass of a motorbike, but generally speaking, in the case of motorbike jumps, the lighter the motorbike (and rider, for that matter!) the better.

- 10 What factors would contribute to the total mass of a motorbike?
- 11 Power-to-weight ratio is calculated by dividing the power output of a vehicle (kW) by the total mass of the motorbike and rider (kg) and multiplying the answer by 1 000 to give units of kW/tonne.
Calculate the power-to-weight ratio of the following motorbikes.
 - a A 250 cc motorbike with power output of 45.4 kW and total mass of 230 kg (bike and rider).
 - b A 500 cc motorbike with power output of 105.9 kW, a total vehicle mass of 280 kg, filled with fuel with a mass of 8 kg and ridden by a woman of mass 95 kg.
- 12
 - a Which of the two motorbikes in question 11 has the higher power-to-weight ratio?
 - b By what percentage is the ratio of this bike greater than that of the other bike?

Stunt flying - dives and speed



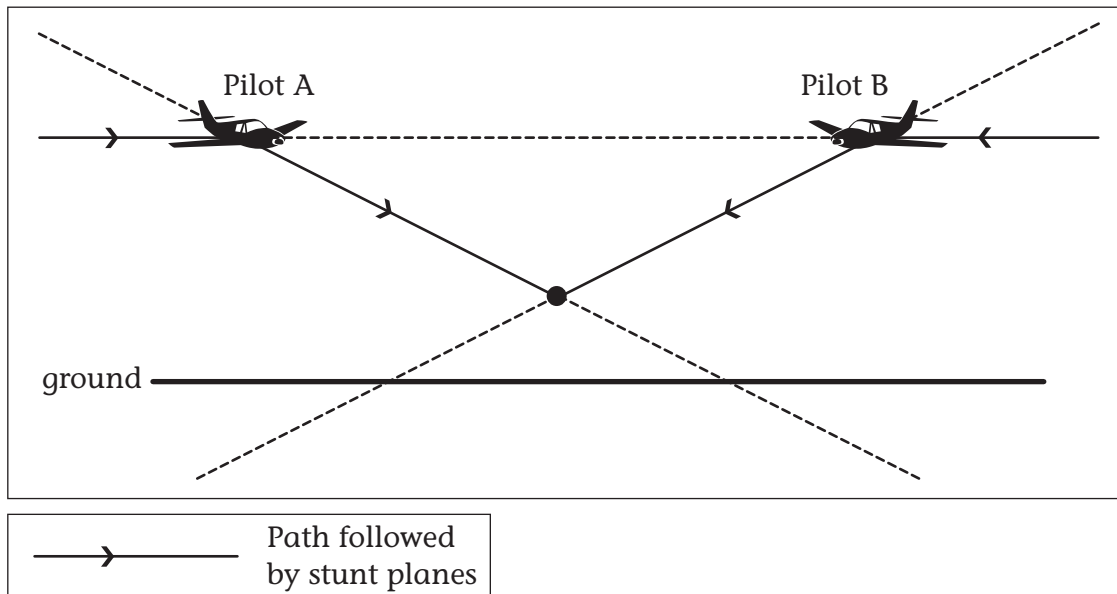
18

Stunt flying is where pilots in a huge range of different planes – from 70 year-old biplanes to modern supersonic jets – perform many different acrobatics in the air.

It is one of the most dangerous and spectacular extreme sports. Pilots must be highly skilled, steady and sure enough to fly at high speeds performing acrobatics and formation flying with absolute precision. They must also enjoy showing off their awesome skills in front of big crowds. Air shows around the world attract millions of people each year, and stunt flying teams are often a part of major sporting and cultural events.

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

Look at this diagram:



Pilot A and Pilot B are both flying in a straight line parallel to the ground, before they enter a dive at an angle of 48° . While this two-dimensional diagram shows the planes on a collision course, they are in fact flying with a horizontal distance of 100 m between their flight paths.

1 Label the diagram to indicate the two 48° angles.

The pilots perform identical dives. Once each pilot reaches the bottom of her dive, she climbs back up in a straight line, level with the starting point of the other plane. So Pilot A is now level with Pilot B's original position (just before the initial 48° dive), and vice-versa.

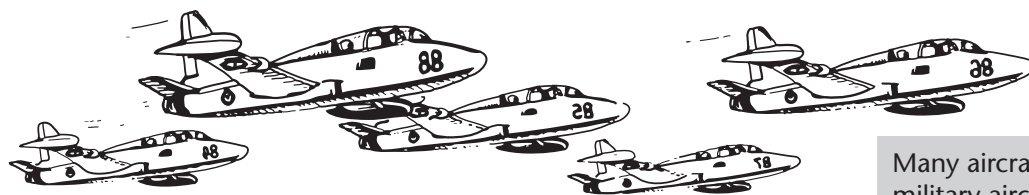
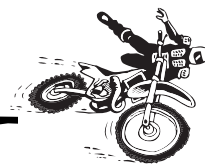
2 At what angle from the ground do the pilots climb back up out of their dives?

3 How many pairs of vertically opposite angles are there on the diagram? Mark the identical pairs on the diagram.

4 The vertical distance between each pilot's original position and the bottom of the dive is 570 metres. Label this distance on the diagram.

5 The distance between the ground and the point marking the bottom of the pilots' dives is 738 metres. What is the original altitude of the pilots?

6 The horizontal distance between the pilots' original positions and the point marking the bottom of the dive is 719 metres. Label this distance on the diagram.



7 What was the actual flying distance of each pilot's dive?

8 If each pilot did a full horizontal rotation when diving and another when climbing back up again, what is the total number of degrees each aeroplane rotated during the stunt?

9 At a recent airshow, an FA 18 Hornet carried out a display flight. At a constant speed of 775 km/h, it rotated through 90 degrees and held this position for 6 seconds, before rotating back to an upright position. Each rotation of 90 degrees took 1 second to complete.

Flying in an upright position for a further 10 seconds, it then rotated through 180 degrees, held this position for 8 seconds, before rotating back to an upright position. Each turn of 180 degrees took 1.5 seconds to complete. It then flew upright for a further 10 seconds, before completing a 360 degree rotation, which took 2.5 seconds.

If the Hornet's speed and direction remained constant for the entire display flight, how far horizontally had the plane flown from the start of the first 90 degree rotation to the end of the final 360 degree rotation?

10 If you could fly in an FA 18 Hornet at a constant speed of Mach 1.8 from Sydney to London, a distance of 17 174 km, how long would it take?

11 A Qantas flight from Sydney to London takes 23 hours and 23 minutes. What is the average flying speed on this flight?

12 Convert the following flying speeds.

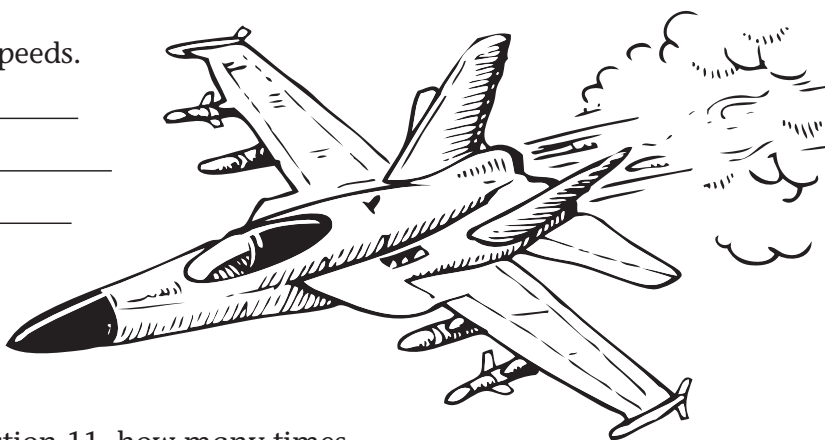
- a Mach 2.1 to km/h _____
- b 2 041 km/h to Mach _____
- c Mach 1.78 to m/s _____

13 An FA 18 Hornet has an average **range** (distance flown with full fuel tanks) of 3 330 km.

Using the information in question 11, how many times would a Hornet need to refuel on a flight from Sydney to London?

14 a A Hornet flies at an average speed of Mach 1.4 for 45 minutes. Calculate the distance in km that it covers in this time.

b Using the average range of 3 330 km, what proportion of fuel is used up on the 45 minute flight? What proportion remains unused?



Many aircraft, particularly military aircraft, can fly in excess of the speed of sound. The speed at which sound travels is equal to 1 225 km/h. This speed is also known as Mach 1.

Australia stages a number of airshows each year in various parts of the country. One type of military aircraft that appears at Australian airshows is the FA 18 Hornet. These aeroplanes can fly at a maximum speed of Mach 1.8, or 1 915 km/h.



In this topic, we'll have a close-up look at some of the specifications of four popular types of aircraft used in stunt flying around the world, including the Pilatus PC-9 which is used by Australia's famous Roulettes. You will need to refer to the information provided about these aircraft on page 41.

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

- 1

a

What is the difference in length in metres between the two biplanes?

b

Now convert your answer to centimetres.
- 2

a

What is the difference in wingspan between the two biplanes?

b

Now convert your answer to centimetres.
- 3

a

What is the difference in height between the two biplanes?

b

Now convert your answer to centimetres.
- 4

On a full tank of fuel, how much further can the Stearman Model 75 travel than the Tiger Moth?
- 5

The difference between each plane's empty mass and its maximum take-off mass is equivalent to the total mass of people and equipment that can be carried in the plane. Complete the table to help you calculate the total mass of people and equipment (the extras) that can be carried in each plane.



Plane	Empty mass (kg)	Max. take-off mass (kg)	Total mass extras (kg)	Total mass extras (t)
Stearman (Boeing) Model 75				
de Havilland DH82 Tiger Moth				
Pilatus PC-9				
Aermacchi MB-326				

- 6

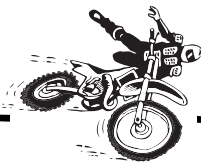
If each plane flew non-stop from Sydney to Adelaide – a flying distance of 1 250 km – and averaged $\frac{3}{4}$ of its maximum speed, how long would the journey take? Complete the table.

Plane	Max. speed (km/h)	$\frac{3}{4}$ of max. speed (km/h)	Time to fly 1 250 km (h)
Stearman (Boeing) Model 75			
de Havilland DH82 Tiger Moth			
Pilatus PC-9			
Aermacchi MB-326			



- 7

As you know, the planes differ in terms of how far they can fly on a full tank of fuel (their range), before they have to land and refuel. Use the information about each plane to identify which ones could fly non-stop from Sydney to Adelaide, and which would have to stop and refuel. For any planes that would need to stop and refuel, state how many times they would need to do this.



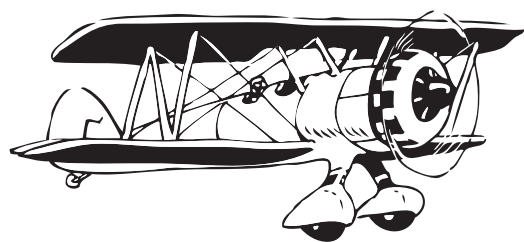
The **range** of an aeroplane refers to how far it can travel when all fuel tanks are full.

The Roulettes – Formed in 1970, the Roulettes are the RAAF’s aerobatic display team. They are often seen at major events, especially sporting and military events, around Australia. Here and overseas, they perform about 150 flying displays every year. The team of five pilots carries out a variety of stunts in flying displays. The Roulettes are based at Sale in Victoria.

Stearman (Boeing) Model 75 - A biplane built in the United States in the 1930s and 1940s. It was built as a military trainer aircraft, but after World War II became popular for crop dusting and competition and stunt flying.

Specifications

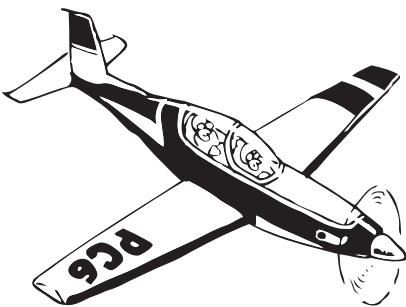
Crew: 2	Maximum take-off mass: 1 232 kg
Length: 7.39 m	Maximum speed: 198 km/h
Wingspan: 9.81 m	Range: 808 km
Height: 2.79 m	
Empty mass: 878 kg	



The Pilatus PC-9 - A turbo-prop aeroplane developed in Switzerland in the early 1980s. It was introduced to Australia for use by the RAAF in 1989. It is also the type of aircraft flown by Australia’s most famous stunt flying team, the Roulettes.

Specifications

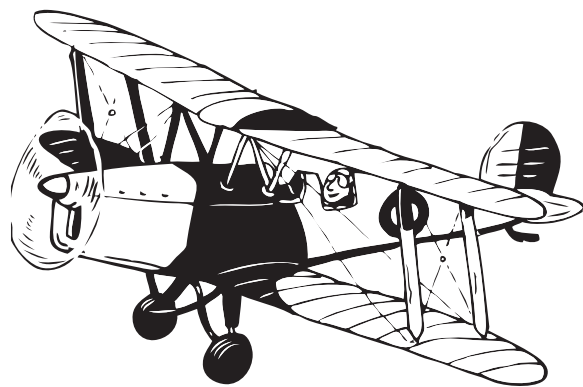
Crew: 1–2	Maximum take-off mass: 3 200 kg
Length: 10.7 m	Maximum speed: 576 km/h
Wingspan: 10.12 m	Range: 1 642 km
Height: 3.26 m	
Empty mass: 1 685 kg	



de Havilland DH82 Tiger Moth - A biplane built in the 1930s and used by Britain’s Royal Air Force as a trainer aircraft until 1952. It is now highly popular as a recreational aircraft and for competition and stunt flying.

Specifications

Crew: 2	Maximum take-off mass: 828 kg
Length: 7.34 m	Maximum speed: 175 km/h
Wingspan: 8.94 m	Range: 486 km
Height: 2.68 m	
Empty mass: 506 kg	

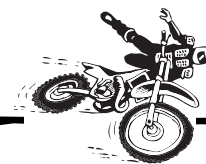


Aermacchi MB-326 - A jet aeroplane developed during the 1950s in Italy. It was designed as a military aircraft, particularly for tasks such as transport and maritime reconnaissance. It has become a popular recreational aircraft and is used for stunt flying. Before the Roulettes acquired their PC-9 aircraft, they used MB-326 aircraft.

Specifications

Crew: 2	Maximum take-off mass: 5 216 kg
Length: 10.65 m	Maximum speed: 871 km/h
Wingspan: 10.85 m	Range: 1 822 km
Height: 3.72 m	
Empty mass: 2 237 kg	





Faster, higher, pushing boundaries further ... Extreme athletes are highly skilled, a bit crazy and very out there. All over the world, extreme sports fanatics are smashing records!

Test your extreme sports knowledge on a few random facts and figures that illustrate some of the awesome, death-defying and downright dangerous stuff that people get up to.

While your answers to many of these questions are likely to come down to best estimates, think about the human feats being performed and base your estimates on a reasonable understanding of what might be possible when someone takes an extreme approach to their sport! You could run a competition in groups, or you might just like to discuss the answers with others. We don't promise there won't be clashes of opinion, but we can at least settle any arguments.

1 Sandboarding is a lot like snowboarding, only it's done on sand dunes instead of snow-covered hills. The longest ever backflip on a sandboard was recorded in 2000 and its length was:

- a** 13.6 m. **b** 10.2 m.
c 18.1 m. **d** 22.5 m.

2 The fastest recorded speed going downhill without the aid of a motor set in 1998 was:

- a** 79 km/h. **b** 94 km/h.
c 101 km/h. **d** 112 km/h.

3 The record in question 2 was achieved by a man:

- a** running. **b** on skis.
c on a horse. **d** on a skateboard.

4 In 1999, Tony Hawk stunned the skating world by landing the first ever:

- a** 900° – 2.5 mid-air rotations.
b 720° – 2 mid-air rotations.
c 1 260° – 3.5 mid-air rotations.
d 1 080° – 3 mid-air rotations.

5 BMX rider Mat Hoffman rode his bike off the edge of a cliff and landed with a parachute, setting a world record. The cliff height was:

- a** 890 m. **b** 940 m.
c 1 020 m. **d** 1 100 m.

6 In 1997, a German bungee jumper set a new record for the length of a jump. He plunged:

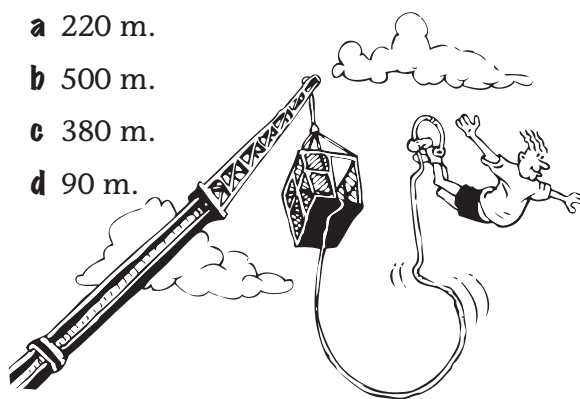
- a** 500 m. **b** 750 m.
c 900 m. **d** 1 000 m.

7 The bungee jumper in question 6 jumped from:

- a** the world's tallest building in Kuala Lumpur.
b a mountain in the Himalayas.
c a helicopter.
d a hot air balloon.

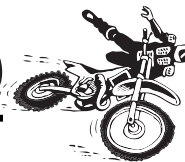
8 The highest fixed bungee jump site in the world available for public use is in Switzerland. Its height is:

- a** 220 m.
b 500 m.
c 380 m.
d 90 m.



9 Street luge is a fast action sport, which has similarities to the luge event in the Winter Olympics. Riders lay down on a wheeled street-luge board and fly down bitumen roads. Typical speeds attained in street luge are:

- a** 75 km/h. **b** 115 km/h.
c 140 km/h. **d** 90 km/h.



- 10** In Hawaii in 1998, Ken Bradshaw surfed the biggest wave ever to be surfed. Its size was approximately:

a 12 m. **b** 25 m.
c 18 m. **d** 8 m.

- 11** The fastest speed ever achieved on a motorcycle was by Chris Carr in 2006. It was:

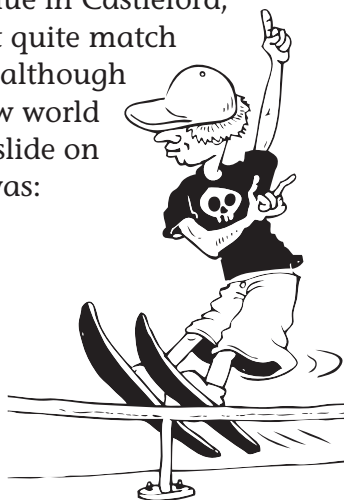
a 563 km/h. **b** 427 km/h.
c 633 km/h. **d** 499 km/h.

- 12** In 2008, two railslide world records were smashed, with both record-breakers from the UK. One of those world records was for snowboard, where Damian turned on an impressive grind of:

a 124.2 m. **b** 58.3 m.
c 44.6 m. **d** 39.3 m.

- 13** At the same venue in Castleford, Charlie couldn't quite match Damian's slide, although he also set a new world record for a railslide on skis. His grind was:

a 36.2 m.
b 41.9 m.
c 56.7 m.
d 118.4 m.



- 14** In 2005, Pascal Bernabe reached a record diving depth on open circuit scuba, which is underwater breathing apparatus where exhaled air all goes into the surrounding water. His dive in the Mediterranean Sea reached:

a 185 m. **b** 330 m.
c 98 m. **d** 264 m.

- 15** The length of time it took Pascal to reach his maximum depth was:

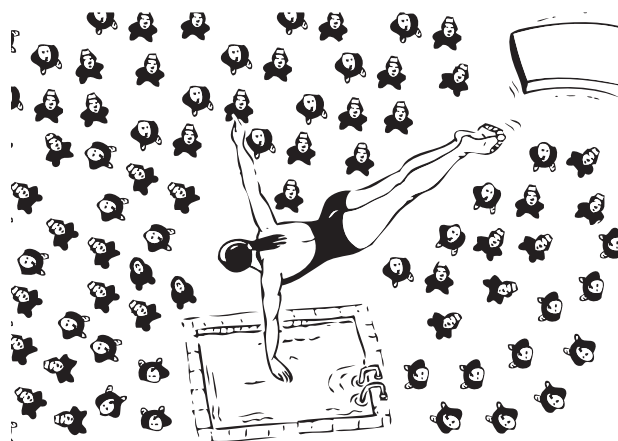
a 1 hour. **b** 30 min.
c 10 min. **d** 2.5 min.

- 16** The length of time it took Pascal to resurface was:

a 8 h 49 min. **b** 2 h 35 min.
c 58 min. **d** 4 h 17 min.

- 17** In 2002, Michael Barber claimed a new world hang gliding record when he flew a distance of:

a 296 km. **b** 412 km.
c 704 km. **d** 828 km.



- 18** The dive into water from the greatest distance ever by a woman was achieved by Lucy Wardle in Hong Kong in 1985. It was from a height of:

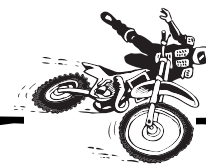
a 18.4 m. **b** 36.8 m.
c 25.2 m. **d** 29.8 m.

- 19** Base jumpers launch themselves off fixed structures – natural or human-made – and land with an open parachute. A Sydney couple, Heather Swan and Glenn Singleman, claimed the world record for a high altitude base jump in 2006 when they jumped off Meru Peak in the Himalayas. Their starting height was:

a 1 070 m. **b** 2 285 m.
c 4 379 m. **d** 6 660 m.

- 20** As base jumpers Swan and Singleman pushed off from the cliff, they very quickly reached free-falling speeds of:

a 65 km/h. **b** 180 km/h.
c 145 km/h. **d** 100 km/h.



Faster, higher, pushing boundaries further ... These extreme sports freaks are addicted to travelling fast – hitting breakneck speeds on road, rail and in the air. Exercise your extreme calculation skills by crunching the numbers involved in these speed records.

Round all your answers to 2 decimal places where appropriate, unless indicated otherwise.

- 1** The land speed record in a vehicle is held by Andy Green, who went an amazing 1 228 km/h in 1997 in his 'ThrustSSC' vehicle powered by two massive jet engines. This is just 7 km/h less than the speed of sound.

a Convert 1 228 km/h to miles per hour (mph).

1 mile = 1 609.34 metres

b Convert this speed to metres per second (m/s).

c What is the speed of sound in m/s?

- 2** The motorcycle land speed record is currently held by Chris Carr, who clocked an awesome speed equivalent to 158.11 m/s.

a Convert this speed to km/h.

b Convert this speed to mph.

The land speed record – maintained for at least 1 km – was first measured in 1898 at a frightening 63.15 km/h. This record was broken many times during the 1900s. Here's a snapshot of some of the landmark records:

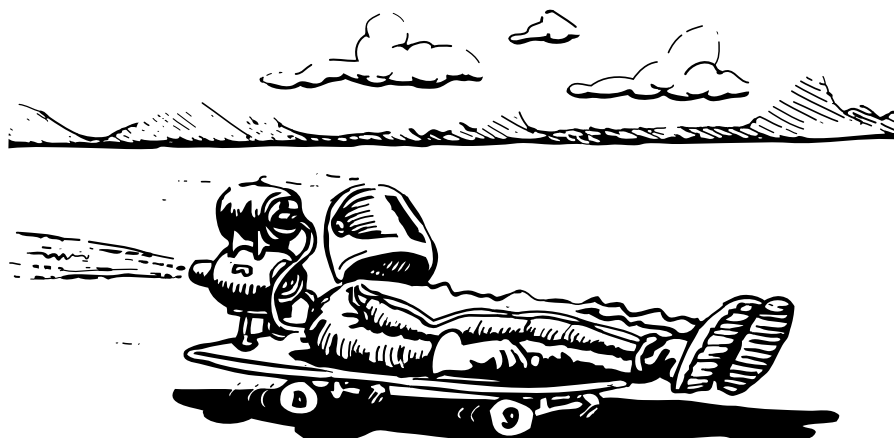
- In 1899, the record stood at a hair-raising 105.88 km/h.
- By 1906, this had increased by nearly 100 km/h to an astonishing 205.44 km/h.
- The 400 km/h mark was exceeded for the first time in 1932, when the somewhat more serious speed of 404.49 km/h was achieved.
- Just 15 years later, the record had cracked the 600 km/h mark – 633.79 km/h in 1947.
- In 1970, the 1 000 km/h barrier was broken, with the record set at 1 014.52 km/h. The record of 1 228 km/h set in 1997 has stood for 11 years, and was the current record at the time of printing this book.

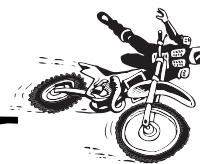
- 3 a** By what factor did the land speed record increase from the previous record in each of the years 1906, 1932, 1947, 1970 and 1997?

b Between which years of those given above did the speeds proportionally increase the most?

c Between which years did the speed increase the most in km/h?

- 4** Construct a column graph to represent the land speed record information. Put each year along the x-axis (horizontal axis) and speed along the y-axis (vertical axis).

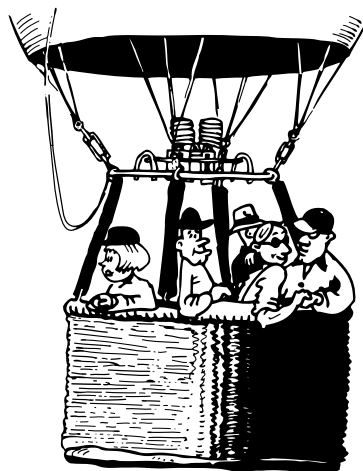




- 5** Not quite sport, but a sporting connection ... In the build-up to the Beijing Olympics in 2008, China set a new world record – it put into service the Beijing-Tianjin Intercity Rail. This high speed train became the fastest commercially operating rail service in the world. At the maximum operating speed on this train, it would be possible to travel 87.5 km in just 15 minutes.
- What is the train's maximum speed in km/h?
 - How far would the train travel in 10 minutes at maximum speed?
 - In travelling time, how long would it take to journey 500 km at this speed?
- 6** On the subject of fast trains, a number of non-commercially operating trains on rails have achieved speeds faster than the Chinese Intercity train. In fact, the highest speed ever recorded by a 'train' – in this case a railed vehicle associated with the testing of US army rocket-propelled missiles – is 10 400 km/h.
- Convert this speed to m/s.
 - How many km/s is this?
 - If a track for a railed vehicle capable of travelling at this speed was built from the east coast across Australia to the west coast – approximately 4 000 km – how long would it take the rocket-propelled vehicle to travel from one side of the country to another if it could maintain this speed?
 - If the vehicle kept travelling at this speed from one side of the country to the other for 2 h 17 min – how many kilometres would it have travelled?

Slowing down a pace now, but still at the extreme end of sports action – hot air ballooning!

- American adventurer and businessman, Steve Fossett, was well known for pushing the boundaries. In 2002, he became the first person to circumnavigate the world non-stop in a hot air balloon. His balloon was 10 storeys high and was named 'The Spirit of Freedom'. He made the trip solo.
- He took off from Northam in Western Australia on 19 June 2002 and returned to land in Queensland on 3 July 2002. The flight took him 13 days, 8 hours and 33 minutes. He covered a distance of 20 626.48 miles.
- Fossett's top speed during the flight was 186 mph over the Indian Ocean, and he set a new record for the longest distance flown in a balloon over 24 hours – 3 186.8 miles on 1 July 2002.
- Steve Fossett went missing in October 2007 after failing to return from flying his plane over the Nevada Desert in the USA. A year later, the wreckage of his plane was found.



- 7**
- For how many hours in total was Fossett flying on his circumnavigation of the Earth?
 - What was Steve Fossett's average speed in mph during the circumnavigation?
 - Convert the total distance flown to km and the average speed to km/h.
 - Convert the maximum speed during the circumnavigation to km/h.
 - If Steve Fossett had flown at his average speed constantly for the entire flight, how far (in km and in miles) would he have flown after:
 - reaching the halfway point?
 - one week?

Answers

1 Conversions - length, time

- 1 a 1 600 b 25.6 c 140 d 0.765
2 a 99.20 b 109.16 c 4 114.80 d 11 876.93 e 21.59 f 68.58
3 a 3.01 b 55.28 c 0.27 d 12.14 e 7.01 f 2.87
4 a 0.45 h b 507 min c 779.85 min = 13.00 h d 2.18 h e 152.40 min = 9 144 s f 2 808 s g 50 880 s h 20
5 a 1 056 min b 17 h 36 min c 17 h 36 min 0 s d 2 780.40 s e 46 min 14.4 s f 34 min 48 s

2 Conversions - speed, mass, volume

- 1 a divide b multiply 2 a 93.17 b 51.52
3 a 3.80 m/s b 5.51 km/h c 937.79 mph d 108.33 m/s
4 a 1 000 b 1 000 c 1 000 000
5 a 531.1 g b 417.3 g c 174.6 kg d 6.53 kg
6 a 7.06 oz b 18.65 lb c 151.78 lb d 42 lb 4 oz
7 23.75 gal 8 2 499 999.97 L ≈ 2 500 000 L
9 97.34 L 10 a 4 000 000 cm³ b 1 055.41 gal

3 Formula One racing

- 1 A street circuit is around streets that already exist in a city, e.g. Melbourne's Formula One circuit at Albert Park. A race circuit is especially built at a particular location to hold car races, e.g. Britain's Formula One circuit.
2 27.78% (~28%) 3 $\frac{16}{18}$ or 0.89 4 16:2 = 8:1
5 3.664 km 6 a 1.710 km b 1.954 km c 0.741 km
7 a tenths, hundredths, thousandths of seconds b 0.885 s c 220.97 km/h d 218.13 km/h e 2.85 km/h slower
8

Team	Driver	Points	Circuit
Wilkins	Rubens Hamilton	3 406	San Marino
Hando	Mark Massa	4 121	Melbourne
Far-rar-ree	Lewis Webber	3 389	Japan
Blue Cow	Felipe Rosberg	3 752	Brazil
McKenzie	Nico Barichello	3 814	China

4 Game fishing - ocean travel

- 1 656 480 W 2 a 35.72 km/h b 34.31 kn
3 a 44.21 km/h b 23.52 kn 4 58.5 km
5 a 50.76 km/h b 1.87 h c 1 h 52 min
6 a 6.46 hours b 6 h 28 min 7 a 69.6 L b \$132.94
8 a \$880 b 0.15 9 Profit = Income - Expenses = \$118.52
10 Profit = 7 040 - 5 167.80 = \$1 872.20

5 Game fishing - catch and release

- 1 Brad, Anastasia, Sally, Omar
2 Anastasia 640 kg, Brad 549 kg, Sally 102 kg, Omar 11 kg
4

Species of fish	Total no. caught	Total mass caught (kg)
Marlin	3	866
Sailfish	4	290
Yellow fin tuna	2	74
Mackerel	9	69

- 7 a 41.29 m total depth b 61.7 m 8 7.91 m/s (≈30 km/h)
9 118.9 kg.m/s

6 Extreme skiing

- 1 a 5.63 min b 4.68 min c 6.87 min
2 a 0.09 h b 0.08 h c 0.11 h
3 Jasmine, 11.11, 2nd; Eli, 12.5, 1st; Nik, 9.09, 3rd
4 a 25.3 km/h b 19.3 km/h c 26.28 km/h
5 a Nik, Jasmine, Eli b The order is different, because the same approximated distance (1 km) was used for each skier in question 3. When you use the actual distances in the calculation, the answers are different. Each skier actually skied more than 1 km, and, because speed = distance ÷ time, the speed increases.
6 There are often obstacles on ski slopes that prevent skiers from skiing straight down, such as moguls, melted snow or ice, trees, rocks, other skiers. Skiing straight down a long slope is likely to cause the skier to build up too much speed, which can lead to a loss of control. Most skiers will ski across slopes as well as down them.
7 a 666 m b 3 952.64 m above sea level
8 a higher b 1 724.64 m
9 3 runs 867.50 AUD, 4 runs 957.83 AUD, 7 runs 1 078.31 AUD

7 Skydiving - falling fast

- 1 1.66 km, 1 660 m 2 1 040 m, 1.04 km
3 a 61.48% b 38.52% 4 5 min 50 s
5 a 8.57% b 91.43% 6 61.48%, 91.43%, 39.52%
7 a 3.25 m/s, b 11.7 km/h 8 188.3 km/h slower.
10 a 0.09 m or 9 cm b 0.11 c below average by 0.02 d higher, by 0.02 e 0.13

8 Sky diving - formations & handicaps

- 1 a 5 points b Lightning (5), Dragonz (4), Blitz (3)
2 a 21 points b Dragonz (19), Blitz (18), Lightning (17)
3 a A handicap system helps balance out the achievement levels for individuals or teams. It enables a team with high levels of skill to compete against others with lower skill levels, and still make meaningful comparisons between their final scores. b golf, horse racing.
4 a Dragonz - 19, Blitz - 19.1, Lightning - 19.04. b Blitz is leading after four rounds.
5 Blitz - 14, Lightning - 15, Dragonz - 17
6

Team	Raw points (without handicap)	Handicap	Final score (with handicap)	Ranking on the day
Blitz	32	1.06	33.92	3
Lightning	32	1.12	35.84	2
Dragonz	36	-	36	1

- 7 a To make a more informed comment on the handicap system, look at the raw scores: Blitz 232, Lightning 245, Dragonz 221. Dragonz scored the least 'raw' points. The results in terms of ranking were the same, whether the handicapping system was used or not. It could be argued that the handicapping system was weighted too heavily towards Blitz and Lightning, as Dragonz were supposedly the best team. In terms of overall rankings, the handicapping system did not affect who finished first, second and third. b No, looking at the points is not enough to make such a judgement. While at the beginning of the season, Dragonz was rated the best team, there could have been a range of reasons why they finished last; e.g. non-availability of regular Dragonz team members; a drop-off in form; improvement in performance of other teams during the season.
8 Each skydiver rotates a total of 1 080°.

Answers (cont'd)

9 V8 Jetsprint racing

- 1 **a** 40 km/h **b** 80 km/h **c** 120 km/h **d** 160 km/h
e 214.4 km/h

2

speed	0	15	30	45	60	75	90	105	120	120	120	120
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- 4 **a** $1\ 200 \times 746 = 895\ 200\text{W}$ **b** 895.2 kilowatts (kW)
5 **a** $850:600 = 17:12$ (or 1.42:1) **b** Power-to-weight ratio is important because the lighter a vehicle is, the faster it can be propelled by its engine.
6 4 – 6, 3 – 8, 8 – 7, 10 – 1, 9 – 8, 3 – 7
7 **a** hundredths **b** 5, 12, 11, 3, 8, 4, 16, 13
8 **a** 25% **b** 7.21 s **c** By 0.89 s **d** 9 **e** 16.86 s
9 **a** 37.5% **b** more, by 12.5%

10 Extreme logic & problem solving

1

Surfer	Beach	Wave height	Month
White	Johanna (Australia)	5.4 m	April
Vermullen	Tamarindo (Costa Rica)	5.2 m	July
Zammut	Jeffreys Bay (South Africa)	6.5 m	March
Telford	North Shore (Oahu, Hawaii)	5.7 m	August
McIntyre	Tavarua (Fiji)	6.8 m	October

2

Team	Rank	Points	Penalties	Total
Hungary	1	1 875	166	1 709
Japan	2	1 648	127	1 521
Chile	3	1 134	106	1 028
Norway	4	1 226	216	1 010

- 3 Japan: vault, Sunday, Chinese judge
China: uneven bars, Saturday, Russian judge
Korea: floor, Tuesday, Austrian judge
USA: balance beam, Monday, Canadian judge

11 White-water rafting - the rapids

- 1 To assist rafters in determining the degree of difficulty likely when paddling a white-water river.
2 So rafters can assess white-water anywhere in the world.
3 Blue River, Red River (easier rapids), Grey River
4 White River, Brown River, Red River (harder rapids)
5 Grade 1: 2; Grade 2: 3; Grade 3: 5; Grade 4: 3, Grade 5: 3
6 1.35 km/h **7** 3 sets of Grade 1, 6 sets of Grade 2
8 0.92 km/h **9** **a** faster **b** 0.43 km/h **c** 430 m/h
10 0.78 km/h
11 Beyond the short course are higher grade rapids, which take longer to navigate, thus slowing them down.
12 Grade 1: 3; Grade 2: 8; Grade 3: 2; Grade 4: 6, Grade 5: 3
13 **a** The 2 km to 3 km stretch. **b** 7

12 White-water rafting - the costs

- 1 $230 + 140 + 90 + 1\ 820 + 40 = \$2\ 320$
2 **a** 78.4% **b** 9.9% **c** 3.9% **3** **a** \$348 **b** \$1 972
4 Hiring = \$2 460, Purchasing = \$2 320. More expensive to hire equipment than buy it.
5 **a** \$819 **b** \$1 001 **c** $\$819 \div \$125 = 6.55$. Seven times.
6 $4.2 \times 2.1 = 8.82\text{ m}^2$, $5.5 \times 2.4 = 13.2\text{ m}^2$ **7** \$4 612
8 $4\ 612 \div 12 = \$384.33$ per rafting trip
9 The total cost for the holiday is \$2 740.

10

Cost	Total \$	Roula's share	Bronwyn's share	Callum's share	Ahmed's share
Raft hire	330.00	-	110.00	110.00	110.00
Equip. hire	360.00 (3 x 120)	-	120.00	120.00	120.00
Fuel	600.00	150.00	150.00	150.00	150.00
Trailer hire	200.00	50.00	50.00	50.00	50.00
Nat. Park entry	50.00	12.50	12.50	12.50	12.50
Camping fees	480.00 (4 x 12 x 10)	120.00	120.00	120.00	120.00
Provisions	720.00	180.00	180.00	180.00	180.00
TOTAL	2740.00	512.50	742.50	742.50	742.50

13 Sphereing or zorbing?

- 1 **a** Bolt: 10.32 m/s. Kolvers: 14.56 m/s. Kolvers was faster. **b** shorter by 91.05 m
2 $2\ 772^\circ$
3 **a** Diameter: the length of a straight line drawn from one point on the circumference to another point, passing through the centre of the circle. **b** $3\text{ m} \div 2 = 1.5\text{ m}$
4 **a** Circumference: the perimeter or distance around the outside of a circle. **b** Surface area: the total 2-D area occupied by the surface of a sphere. **c** The volume of a sphere (or any other 3-D shape) refers to the total 3-D space occupied by the shape.
5 **a** 1.46 m **b** 9.14 m **c** 1.96 m **d** 23 cm
6 **a** Outer sphere: 26.60 m^2 **b** Inner sphere: 12.07 m^2
7 **a** 1. Calculate the volume of the outer sphere.
2. Calculate the volume of the inner sphere.
3. Subtract the inner sphere volume from the outer sphere volume. **b** 8.96 m^3
8 **a** 0.98 m **b** 13.03 m^3 **c** 3.94 m^3 **d** 9.09 m^3
9 **a** $7\ 055\text{ cm}^3$ **b** 1 846

14 Skateboarding - jump, grind & slalom

- 1 **a** Length 24.49 m; Height 7.29 m **b** 50%
c Length 158 ft or 48.98 m, Height 47 ft or 14.57 m
d e.g. the angle at which the jump is built; the resistance experienced through the air; the size and mass of the skateboard; size and mass of the skater
2 $10.31 \div 15.00 = 0.69$ **3** **a** $\frac{69}{100}$ **b** 69% **4** 1.32 m
5 It is possible to attain much higher speeds on skis and snowboards before jumping on to the grind rail, therefore the length of grinds possible are far greater.
6 **a** 3.57 s **b** 14.13% **c** every 0.22 s
7 **a** J. Nguyen 26.53 s, R. Sampson 23.60 s, W. Forsyth 22.92 s, Y. Ibrahim 27.42 s
b 1. Forsyth, 2. Sampson, 3. Nguyen, 4. Ibrahim
8 **b** 760 mm, 76 cm **c** 37 cm

15 Skateboarding - decks, trucks & wheels

- 1 **a** cm = feet $\times 30.48$ or 1 ft = 30.48 cm
b cm = inches $\times 2.54$ or 1 in = 2.54 cm
2 **a** Samantha: 83:50, Adrian: 9:4 **b** Adrian's
3 $172.72\text{ cm} \div 81.28\text{ cm} = 2.13:1 = 213:100$
4 The taller a person is, the greater the recommended ratio of height to deck length.
5 e.g. the taller a person is, the heavier they are also likely to be. A higher height:length ratio means there is more deck area across which to spread the mass of taller and heavier skateboarders. The taller a person is, the longer their legs and stride are likely to be, and the more deck

Answers (cont'd)

- length they are likely to need in order to be comfortable.
- 6 Wheels with a larger diameter will roll faster than those with a smaller diameter, because each turn of the larger wheel results in more distance being covered compared with the smaller wheel.
- 7 Speed is important for skateboarders who ride ramps, and wheels of larger diameter will roll faster.
- 8 a 175.93 mm b C = 17.59 cm 9 2.82 cm
- 10 a Having more wheel surface in contact with the ground means more grip on the road, which could be helpful when cornering. b More wheel surface in contact with the ground means the skateboard will travel slower due to greater friction.
- 11 67.82 cm²

16 Luge

- 1 a 1.463 km b 0.917 km c 0.546 km
- 3 The men's course was the steepest.
- 4 a Women's course 6.42% b Men's course 7.76%
- 5 ratio, length, percentage, higher
- 6 a Woman 30.64 m/s, man 35.87 m/s
b Woman 110.29 km/h, man 129.14 km/h
- 7 Single luge 23 kg, Double luge 27 kg 8 57.22 m
- 9 191.49 times 10 49 895 ÷ 1 200 = 41.58 = 42
- 11 a In this position, there is minimal air resistance – the body is low to the ground. If a competitor knelt on the luge, there would be much more air resistance because the body is upright and more exposed, which keeps the speed down. b Answers could include: it's uncomfortable; the head is placed at the rear so vision is poorer – the rider has to lift the head to see ahead of the luge.

17 An insane record motorbike jump

- 1 322 ft × 0.31 = 99.82 m
- 2 a 55 ft b 13.95 m
- 3 Length of jump: 322 ft, 99.82 m.
Maximum height: 60 ft, 18.60 m.
Exit speed: 94 mph, 151.34 km/h
- 4 a See table at right.
b 0.38 : 1 = 19 : 50
c Yes, this ratio was constant.
Mathematical explanation: because the rate at which the height and distance changed during the ascent was constant in both cases, the ratio of one to the other remains constant also. In terms of the feat performed: a jump like this goes along the trajectory of an arc, which means the horizontal distance and vertical distance remain proportional to one another on the way up, until the halfway point when they become inversely proportional to one another.
d Because if the height of the jump is too great, the ratio increases, and the rider sacrifices distance. If the height is too small, the ratio decreases, and distance is sacrificed.
- 6 The same, because the ratio between jump height and distance is the same.
- 7 Cubic centimetres, a measurement of the total volume of combined space that an engine's pistons displace as they move from the top of the cylinder to the bottom of the cylinder.
- 8 a 2 : 1 b 111 : 100 c No, because a number of other factors come into play in performing jumps like this, not just the engine size.

Jump height (m)	Jump distance (m)
0	0
2.33	6.2
4.65	12.4
6.98	18.6
9.3	24.8
11.63	31
13.95	37.2
16.28	43.4
18.6	49.6
16.28	55.8
13.99	62
11.63	68.2
9.3	74.4
6.98	80.6
4.65	86.8
2.33	93
0	99.2

- 9 e.g. the speed at which the rider exits the ramp; the shape of the ramp; the mass of the motorbike and rider; the atmospheric conditions such as presence of wind; how the rider controls the motorbike in the air.
- 10 e.g. mass of the motorbike; mass of the rider; amount of fuel in the tank
- 11 a 197.39 kW/t b 276.5 kW/t 12 a 500 cc b 40.07%

18 Stunt flying - dives and speed

- 2 48° 3 10 pairs 5 1 308 metres 7 917.53 m 8 720°
- 9 7.75 km 10 8.97 hours 11 734.56 km/h
- 12 a 2572.5 km/h b Mach 1.67 c 605.7 m/s
- 13 5.15. The Hornet needs to refuel 6 times, including the initial filling up of the tanks at the start.
- 14 a 1 286.25 km b 0.61 has been used, 0.39 is left.

19 Stunt flying - the aircraft

- 1 a 0.05 m b 5 cm 2 a 0.87 m b 87 cm
- 3 a 0.11 m b 11 cm 4 322 km
- 5

Plane	Empty mass (kg)	Max. take-off mass (kg)	Total mass extras (kg)	Total mass extras (t)
Stearman (Boeing) Model 75	878	1232	354	0.354
de Havilland DH82 Tiger Moth	506	828	322	0.322
Pilatus PC-9	1 685	3 200	1 515	1.515
Aermacchi MB-326	2 237	5 216	2 979	2.979

6

Plane	Max. speed (km/h)	¾ of max. speed (km/h)	Time to fly 1250 km (h)
Stearman (Boeing) Model 75	198	148.5	8.42
de Havilland DH82 Tiger Moth	175	131.25	9.52
Pilatus PC-9	576	432	2.89
Aermacchi MB-326	871	653.25	1.91

- 7 Stearman Model 75 (once) and deHavilland Tiger Moth (twice) would need to refuel. The Pilatus and the Aermacchi planes would not.

20 For the record - people, power and nature

- 1 a 2 c 3 b 4 a 5 d 6 d 7 c 8 a 9 b 10 b 11 a
- 12 d 13 a 14 b 15 c 16 a 17 c 18 b 19 d 20 d

21 For the record - engines and speed

- 1 a 762.73 mph b 341.14 m/s c 343.06 m/s
- 2 a 569.20 km/h b 353.54 mph
- 3 a 1899–1906: a factor of 1.94.
1906–32: a factor of 1.97. 1932–47: a factor of 1.57.
1947–70: a factor of 1.6. 1970–97: a factor of 1.21.
b Between 1906 and 1932.
c Between 1947 and 1970 (380.73 km/h).
- 5 a 350 km/h b 59.5 km c 1.43 h (1 h 26 min.)
- 6 a 2 888.89 m/s b 2.89 km/s c 0.38 h = 22.8 min
d 23 712 km
- 7 a 320.55 h b 64.35 mph c Distance flown
33 208.63 km, av. speed 103.60 km/h d 299.46 km/h
e Halfway point: 16 604.31 km, 10 313.24 miles.
1 week: 17 401.32 km, 10 808.28 miles

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